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EEA108D-E06
Course: EEA108D Power System Design - Capstone Project
Group 1 - Commercial Building Electrical Power System
School of EECE, Mapua University

1 Title Page

Design and Engineering Evaluation of the Electrical Power System of a Commercial Building

FEEDER AND TRANSFORMER SIZING

Power factor correction and commercial-load characteristics

Engineering Design Report | Revision 0 | 10 October 2026

Six-storey mixed-use commercial building, Metro Manila, Philippines. Design boundary: one utility service, transformer station, main distribution board, and floor/area panels.

Recommended design: one 1,250 kVA transformer, 2,500 A main board, 400 kVAr detuned automatic capacitor bank, and 400 kVA essential standby generator. All numerical results are from the stated assumptions and the supplied independent Python solver.

Design status: academic engineering design using assumed utility and equipment data. Protection settings are preliminary; manufacturer time-current curves, actual utility limits, and approved equipment submittals are commissioning inputs. No professional seal or construction approval is represented.

2 Project Description

The project develops the electrical supply and distribution subsystem of a 12,000 m² commercial building. Two retail floors and four office floors share the normal supply. The scope includes utility intake, transformers, cables, switchgear, earthing, essential standby supply and automatic power-factor correction. Architectural, structural, mechanical equipment design and final branch-circuit layouts are outside the boundary.

The attached EEA108D capstone assignment establishes Group 1, the feeder/transformer focus, the required analyses, and report structure [1]. The designer supplies the missing brief. Quantities below are planning assumptions, not a site survey or utility offer.

Table 1. Document and model controls

Item	Control
Author group	JOHN RAY O. DALIDA, KENT ALDRICH CUSI, REYNHARD HEINRICH PASION
Course / group	EEA108D-E06 / Group 1
Date / revision	10 October 2026 / 0
Simulation engine	Custom Python + NumPy backward/forward sweep; nodal impedance fault check
Power convention	Positive kW and kVAr are load consumption; capacitors inject kVAr
Study years	Year 0 commissioning basis; Year 10 planning limit
Accuracy	Unrounded data in JSON/CSV; displayed results rounded to 2-4 decimals
Frozen input checksum	4af2946ec053642f25383ae517897351 57014c8b6aba06a59aaea27147967012

Reading the numerical labels

Assumed identifies a designer input. **Reference** identifies a published methodological or tabulated basis. **Calculated** means arithmetic from those inputs. Selected ratings are design outputs. No manufacturer quotation or utility tariff is invented.

3 Design Objectives

Table 2. Objective-to-evidence register

No.	Objective	Evidence
1	Determine power requirements	Section 5: connected and coincident load; growth table
2	Develop configuration	Section 6 and drawings E-01/E-03
3	Rate equipment	Section 7: transformer, feeder, breaker and generator calculations
4	Evaluate voltage and loading	Section 8 and independent load-flow tables
5	Determine fault levels	Section 9: source, transformer, cable and motor bound
6	Develop protection	Section 10 and E-04
7	Plan expansion	Section 11: year-by-year checks and spare capacity
8	Compare feasible alternatives	Sections 13-14: one unit vs split-bus pair
9	Verify calculations	Section 18: executed independent solver and error comparison
10	Prepare report and final SLD	This report, E-02 and separate PNG drawing set

Acceptance criteria are applied to present and future operation. A normal-load pass does not imply full-building standby coverage or full N-1 redundancy. Fire operation, APFC unavailability and transformer-outage behavior are identified separately.

4 Design Requirements and Assumptions

Step 0 - Design Brief (Assumed by the Designer)

Every numerical entry in Tables 3-7 is **Assumed**, including values selected to align with published standards. This brief is frozen before calculation. Later cable, breaker and equipment selections are design outputs; they do not alter the load inputs. General standards give methods, not a universal office/retail demand factor.

Table 3. Design Brief (Assumed by the Designer)

Input	Frozen assumed value	Basis / rationale
Facility	Assumed: 6 floors × 2,000 m² = 12,000 m²; Metro Manila	Modest mid-rise commercial building; designer scenario
Use mix	Assumed: floors 1-2 retail, 4,000 m²; floors 3-6 office, 8,000 m²	No residential or industrial occupancy
Utility	Assumed: 13.8 kV, 3-phase, 60 Hz; 1 utility feeder; 250 MVA maximum initial fault level; X/R = 10	Local-style MV service; confirm with distribution utility [2]
Utility operating range	Assumed: 0.975-1.025 pu at source for study	Project connection requirement, not a guaranteed PDC range
HV route	Assumed: 20 m, 3 × 50 mm² Cu shielded; R = 0.494 Ω/km, X = 0.120 Ω/km	Conservative planning cable inputs; verify submittal
LV distribution	Assumed: 400/230 V, 3-phase, 4-wire; separate PE; TN-S downstream	IEC-style equipment base. Exact phase voltage is 400/√3 = 230.94 V [5]
Transformer alternatives	Assumed: A1 1 × 1,250 kVA; A2 2 × 800 kVA; both Dyn11, 6% Z	Standard procurement candidates; LV tie normally open for A2 [6]
Transformer loss model	Assumed A1: r = 1.0%, P0 = 1.8 kW; A2 each: r = 1.1%, P0 = 1.3 kW	Guaranteed-loss assumptions for comparison, not a vendor quote
Growth / reserve	Assumed: 3%/year, 10 years, 10% design margin; 0.95 harmonic capacity factor	Planning assumptions; no step change in occupancy
Power factor	Assumed planning initial 0.88 lagging; target PCC 0.97 displacement, minimum 0.95 true PF	Actual initial aggregate will be calculated from loads
Voltage criteria	Assumed: load 0.95-1.05 pu; feeder drop ≤3%; wiring total ≤5%; branch reserve 1%	Project criteria aligned with good practice [3,5,10]
Physical / budget	Assumed: electrical plant area allowance 60 m²; PHP 25,000,000 soft target; 30% spare tray and 25% spare ways	Electrical space allocation only. Estimate must expose overruns
Standby	Assumed: 400 kVA / 320 kW, 400 V, 60 Hz; essential supply only; no utility parallel	Fire pump, one lift, one water pump, IT/UPS and life safety
Hours	Assumed: offices 08:00-18:00; retail 10:00-22:00; IT/security 24 h/day	Daily profile independently specified below

Step 0 continued - frozen initial load schedule

Table 4. Design Brief load inputs - all numeric cells Assumed

Load	Qty × unit rating	η	PF	Demand factor	Coincidence
General LED lighting	Assumed 3000 × 0.04 kW	Assumed 1	Assumed 0.95	Assumed 1	Assumed 1
General receptacle allowance	Assumed 1200 × 0.15 kW	Assumed 1	Assumed 0.9	Assumed 0.6	Assumed 1
VRF outdoor units (electrical input)	Assumed 36 × 12 kW	Assumed 1	Assumed 0.85	Assumed 0.85	Assumed 0.9
Elevators (shaft rating)	Assumed 2 × 18.5 kW	Assumed 0.9	Assumed 0.82	Assumed 1	Assumed 0.65
Water pumps, one duty/one standby (shaft)	Assumed 2 × 7.5 kW	Assumed 0.9	Assumed 0.82	Assumed 1	Assumed 0.5
Electric fire pump (shaft), fire case only	Assumed 1 × 37 kW	Assumed 0.9	Assumed 0.85	Assumed 0	Assumed 1
IT/UPS input, includes UPS losses	Assumed 2 × 30 kW	Assumed 1	Assumed 0.95	Assumed 0.9	Assumed 1
Escalators (shaft rating)	Assumed 2 × 7.5 kW	Assumed 0.9	Assumed 0.82	Assumed 0.75	Assumed 1
Emergency lighting, controls, security	Assumed 1 × 20 kW	Assumed 1	Assumed 0.95	Assumed 1	Assumed 1
Tenant appliances, excludes REC/HVAC/IT	Assumed 12 × 10 kW	Assumed 1	Assumed 0.85	Assumed 0.65	Assumed 1

Motor shaft ratings are divided by the assumed efficiency. VRF ratings are already electrical input. The IT rating includes UPS conversion losses; it is not added again. General receptacles exclude the separately listed IT and tenant appliances. Fire-pump normal demand factor is zero; fire operation uses its full input in a separate case.

Assumed connected lighting density is 10 W/m² (3,000 luminaires at 40 W). This is below the Philippine Green Building Code office benchmark of 10.8 W/m² [9]; retail lighting remains subject to the applicable space-by-space limit. Assumed receptacle allowance is 15 W/m² and VRF electrical input is 36 W/m². These are designer planning allowances, not PEC minimum load prescriptions. Selected demand factors 0.60-1.00 and coincidence factors 0.50-1.00 reflect the specified operating pattern using the utilization/coincidence method [8].

Table 5. Design Brief route lengths - all numeric cells Assumed

Circuit	Assumed length	Circuit	Assumed length
Transformer to MDB	15 m	F1 / F2	25 / 35 m
F3 / F4	45 / 55 m	F5 / F6	65 / 75 m
EDB / MDP	30 / 60 m	Dedicated fire pump	35 m
APFC / tie	5 / 10 m	Generator to emergency bus	20 m

Step 0 continued - methods, harmonics and operating constraints

Table 6. Design Brief calculation assumptions

Category	Frozen assumed values / application
Cable installation	Assumed XLPE Cu single cores, ventilated/separated tray tiers; conservative 70°C method-C tabulated baseline; ambient 40°C; kT=0.87, kG=0.80, kN=0.86. Grouping factor requires no more than two touching circuits per tier, with separation between tiers.
Cable impedances	Assumed R = 23.7/A Ω/km at conservative hot condition; R20 = 17.241/A Ω/km for maximum faults; X = 0.096 Ω/km at 60 Hz. These simplified values follow [10,11]; equal-length parallel sets use Z/n.
Harmonics	Assumed operating THDi=15% of fundamental; design envelope 20% triplen content, 200% neutral area for floors/EDB/main; no harmonic spectrum claimed.
Fault method	Assumed cmax=1.10, cmin=0.95, minimum transformer Z=0.90 × nominal; motor bound = 4 × all connected future motor rated current; motor X/R=4.
Generator	Assumed x″d=0.15 pu, X/R=10; DOL locked rotor = 6 × motor FLC; largest active-power pickup step ≤25% of rated kW; assumed transient motor-start dip limit 15%.
APFC	Assumed maximum 400 kVAr net, 25 kVAr smallest aggregate step; reactor p=7%; 480 V capacitor elements; losses 5 W/kVAr switched; 30 s step delay; ≥60 s discharge/reclose delay.
Earthing	Assumed soil 100 Ωm; rods 3 m ×16 mm; project electrode target ≤5 Ω; TN-S metallic PE carries faults, not the soil.
Economics	Assumed energy PHP 12/kWh; recognized PF penalty PHP 0/year; 15% installation +15% contingency on installed base; ±30% estimate uncertainty; 8% real discount, 20-year evaluation.
Reliability / maintenance	Assumed essential loads retained on generator; ordinary retail and office loads may be shed. A2 retains 55% normal demand on transformer N-1. Assumed annual maintenance allowance 1% of materials.

Table 7. Design Brief daily operating profile - all numeric entries Assumed

Time block	Hours	Fraction of peak kW	Planning displacement PF
00:00-06:00	6	0.2	0.85
06:00-08:00	2	0.45	0.86
08:00-12:00	4	0.85	0.89
12:00-16:00	4	1	0.886
16:00-18:00	2	0.85	0.88
18:00-22:00	4	0.6	0.86
22:00-24:00	2	0.25	0.82

The detailed load sum governs the peak PF; the rounded 0.886 daily-profile value is only a planning indicator. No daily energy shape is represented as measured data. Fire tests are not included in annual energy.

Standards and design hierarchy

Table 8. Applicable references and limits of use

Reference	Use in this design
PEC 2017 [3]	Local installation, conductor and overcurrent requirements; continuous loads, motors, standby and fire-pump treatment. No unverified clause numbers are asserted.
Philippine Distribution Code 2017 [2]	Connection, metering, power quality and coordination with the distribution utility. Actual offered voltage/fault data override assumptions.
IEEE 141 / 242 / 399 [4]	Distribution, protection and study methodology. These are legacy Color Books, not a claim of current mandatory editions.
IEC 60909 [11,12]	Equivalent-voltage source fault method, impedance contributions and maximum/minimum study concepts. Motor contribution is deliberately a bounding simplification.
IEC 60076 / IEEE C57 [6,13]	Transformer specification, impedance, losses, connection, temperature and loading requirements.
IEC 60364 [5,7,10]	Cable thermal sizing, voltage drop, earthing and installation reference methods. PEC has local precedence where requirements differ.
Manufacturer guides [7,8,10,11,14]	Public technical basis for ampacity, utilization, voltage drop, fault method and detuned capacitor equipment.

PEC minimum branch-circuit connected loads and actual terminal temperature ratings must be reconciled in the permit design. This report does not claim that an IEC cable table by itself proves PEC compliance. Conservative ampacity, full neutral provision and explicit continuous-load margins are applied here. Published standards were identified through official or manufacturer sources; complete copyrighted standards are not reproduced.

5 Load Assessment

Task 1 - connected and coincident demand

```
Pconnected = quantity × unit kW / η
Pdemand = Pconnected × demand factor × coincidence factor
Qdemand = Pdemand × tan(arccos(pf))
Smaximum = √[(ΣP)² + (ΣQ)²]; pfaggregate = ΣP / Smaximum
```

Table 9. Calculated load schedule

Load	Connected kW	DF	CF	Demand kW	PF	kVAr
LGT	120.00	1	1	120.00	0.95	39.44
REC	180.00	0.6	1	108.00	0.9	52.31
HVAC	432.00	0.85	0.9	330.48	0.85	204.81
LIFT	41.11	1	0.65	26.72	0.82	18.65
WP	16.67	1	0.5	8.33	0.82	5.82
FP	41.11	0	1	0.00	0.85	0.00
IT	60.00	0.9	1	54.00	0.95	17.75
ESC	16.67	0.75	1	12.50	0.82	8.73
EM	20.00	1	1	20.00	0.95	6.57
TEN	120.00	0.65	1	78.00	0.85	48.34
TOTAL	1,047.56	-	-	758.04	0.8833	402.42

```
ΣPconnected = 120.00 + 180.00 + 432.00 + 41.11 + 16.67 + 41.11 + 60.00 + 16.67 + 20.00 + 120.00
= 1,047.56 kW
ΣPdemand = 758.04 kW; ΣQdemand = 402.42 kVAr
S0 = √(758.0356² + 402.4187²) = 858.2300 kVA
pf0 = 758.0356/858.2300 = 0.883255 lagging
Building demand factor = 758.0356/1047.5556 = 0.7236
```

No additional global diversity is applied after the row factors. Coincidence is ≤1; the corresponding diversity is its reciprocal. This avoids applying diversity twice. Normal fire-pump demand is zero but connected installed power includes the fire pump. Status: **OK**, normal and fire cases are distinct.

Task 1 - every load-row calculation

LGT - General LED lighting

```
Pconn = 3000 × 0.0400/1.00 = 120.0000 kW
Pd = 120.0000 × 1.00 × 1.00 = 120.0000 kW
Qd = 120.0000 × tan(arccos 0.95) = 39.4421 kVAr
```

REC - General receptacle allowance

```
Pconn = 1200 × 0.1500/1.00 = 180.0000 kW
Pd = 180.0000 × 0.60 × 1.00 = 108.0000 kW
Qd = 108.0000 × tan(arccos 0.90) = 52.3068 kVAr
```

HVAC - VRF outdoor units (electrical input)

```
Pconn = 36 × 12.0000/1.00 = 432.0000 kW
Pd = 432.0000 × 0.85 × 0.90 = 330.4800 kW
Qd = 330.4800 × tan(arccos 0.85) = 204.8131 kVAr
```

LIFT - Elevators (shaft rating)

```
Pconn = 2 × 18.5000/0.90 = 41.1111 kW
Pd = 41.1111 × 1.00 × 0.65 = 26.7222 kW
Qd = 26.7222 × tan(arccos 0.82) = 18.6522 kVAr
```

WP - Water pumps, one duty/one standby (shaft)

```
Pconn = 2 × 7.5000/0.90 = 16.6667 kW
Pd = 16.6667 × 1.00 × 0.50 = 8.3333 kW
Qd = 8.3333 × tan(arccos 0.82) = 5.8167 kVAr
```

FP - Electric fire pump (shaft), fire case only

```
Pconn = 1 × 37.0000/0.90 = 41.1111 kW
Pd = 41.1111 × 0.00 × 1.00 = 0.0000 kW
Qd = 0.0000 × tan(arccos 0.85) = 0.0000 kVAr
```

IT - IT/UPS input, includes UPS losses

```
Pconn = 2 × 30.0000/1.00 = 60.0000 kW
Pd = 60.0000 × 0.90 × 1.00 = 54.0000 kW
Qd = 54.0000 × tan(arccos 0.95) = 17.7489 kVAr
```

ESC - Escalators (shaft rating)

```
Pconn = 2 × 7.5000/0.90 = 16.6667 kW
Pd = 16.6667 × 0.75 × 1.00 = 12.5000 kW
Qd = 12.5000 × tan(arccos 0.82) = 8.7251 kVAr
```

EM - Emergency lighting, controls, security

```
Pconn = 1 × 20.0000/1.00 = 20.0000 kW
Pd = 20.0000 × 1.00 × 1.00 = 20.0000 kW
Qd = 20.0000 × tan(arccos 0.95) = 6.5737 kVAr
```

TEN - Tenant appliances, excludes REC/HVAC/IT

```
Pconn = 12 × 10.0000/1.00 = 120.0000 kW
Pd = 120.0000 × 0.65 × 1.00 = 78.0000 kW
Qd = 78.0000 × tan(arccos 0.85) = 48.3401 kVAr
```

Task 1 - allocation and load growth

Lighting, receptacles, VRF and tenant demand are divided equally among six floors. The two escalators are split between the two retail floors. IT and emergency loads are assigned to EDB; lifts and duty water pumping are assigned to MDP. Local essential transfers select one lift and one water pump during outage, so their loads are not duplicated in EDB.

```
Pfloor base = (120 + 108 + 330.48 + 78)/6 = 106.08 kW
Pretail floor = 106.08 + 12.5/2 = 112.33 kW
PEDB = 54 + 20 = 74 kW
PMDP = 26.7222 + 8.3333 = 35.0556 kW
Pn = P0(1+g)^n; Qn = Q0(1+g)^n
P10 = 758.035556(1.03)^10 = 1018.736399 kW
Q10 = 402.418653(1.03)^10 = 540.817019 kVAr
S10,uncorrected = 858.229967 × 1.343916379 = 1153.389310 kVA
```

Table 10. Calculated ten-year growth, no further diversity

Year	Substitution factor (1.03)^n	kW	kVAr	Uncorrected kVA
0	1.000000	758.04	402.42	858.23
1	1.030000	780.78	414.49	883.98
2	1.060900	804.20	426.93	910.50
3	1.092727	828.33	439.73	937.81
4	1.125509	853.18	452.93	965.95
5	1.159274	878.77	466.51	994.92
6	1.194052	905.13	480.51	1,024.77
7	1.229874	932.29	494.92	1,055.51
8	1.266770	960.26	509.77	1,087.18
9	1.304773	989.06	525.07	1,119.80
10	1.343916	1,018.74	540.82	1,153.39

The design allows 34.39% aggregate growth. Normal connected load is not a maximum-demand measurement. The initial uncorrected aggregate is 68.66% of the candidate transformer rating before network losses; final loading comes from the solver.

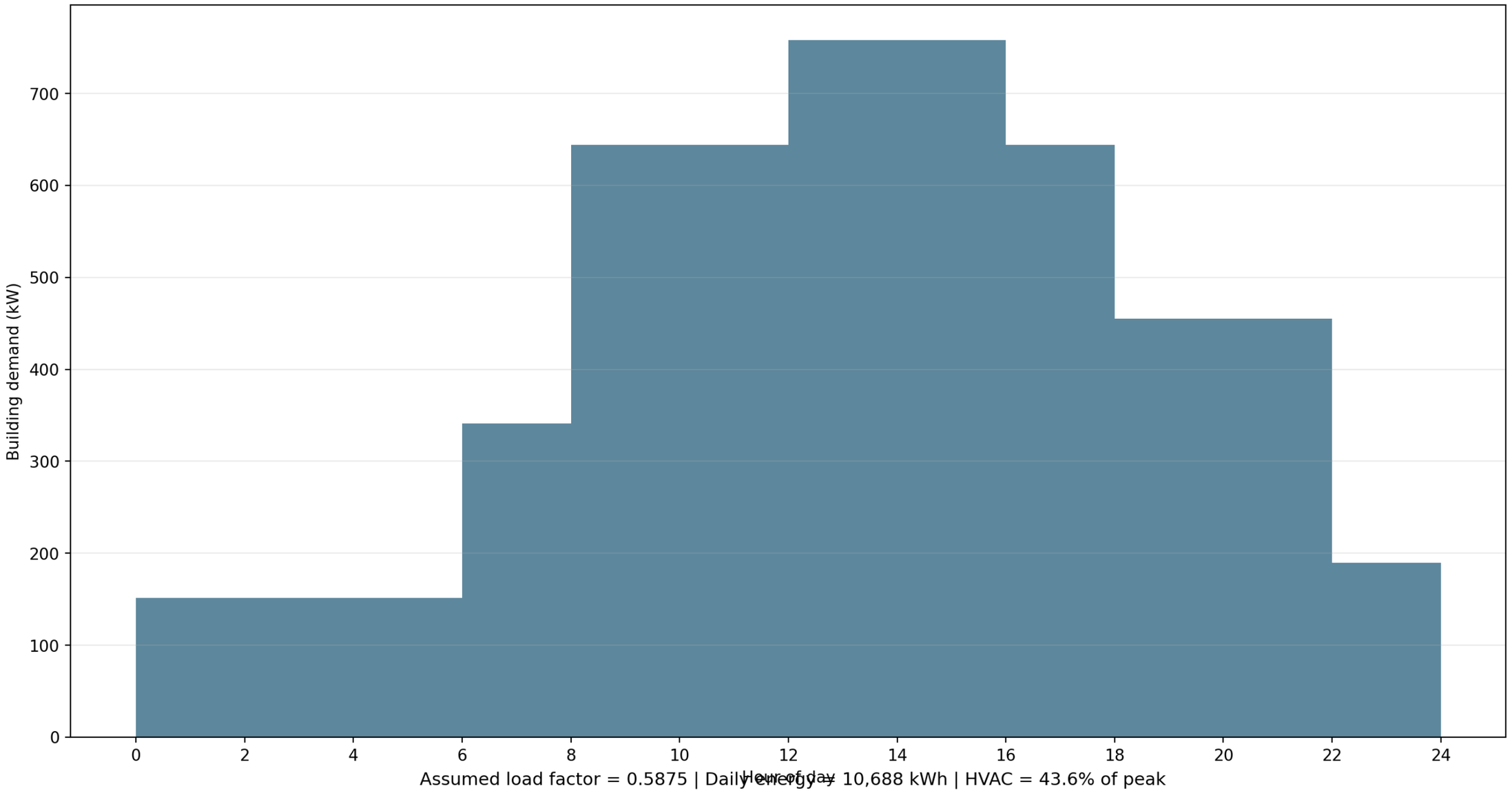
Task 1 - commercial operating characteristics

```
Daily energy = 758.0356 × [6(0.20)+2(0.45)+4(0.85)+4(1.00)
+2(0.85)+4(0.60)+2(0.25)]
= 758.0356 × 14.10 = 10688.3013 kWh/day
Load factor = 14.10/24 = 0.5875
Annual load energy = 10688.3013 × 365
= 3901229.99 kWh/year
Loss factor = [6(.20²)+2(.45²)+4(.85²)+4(1²)
+2(.85²)+4(.60²)+2(.25²)]/24 = 0.439375
HVAC peak share = 330.48/758.0356 ×100 = 43.60%
```

The assumed weekday-shaped profile is applied to all 365 days for a transparent annual screening estimate; actual weekends and holidays would reduce energy. HVAC drives the daytime peak. Retail extends the shoulder into the evening. IT/security produces a continuous base load. Lifts, pumps and escalators are intermittent; common-area and retail schedules are partly coincident.

LED drivers, UPS rectifiers and VRF/VFD inputs are nonlinear. Capacitors correct fundamental reactive demand, not distortion power. The study therefore keeps a 15% THDi current allowance, uses substantial neutral capacity, and specifies detuned APFC. The model is balanced; phase-unbalance and a measured harmonic spectrum remain verification requirements before final procurement.

Commercial daily load profile | assumed planning day



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Figure 1. E-09: assumed commercial daily profile; all energy is calculated from this profile.

6 Proposed System Configuration

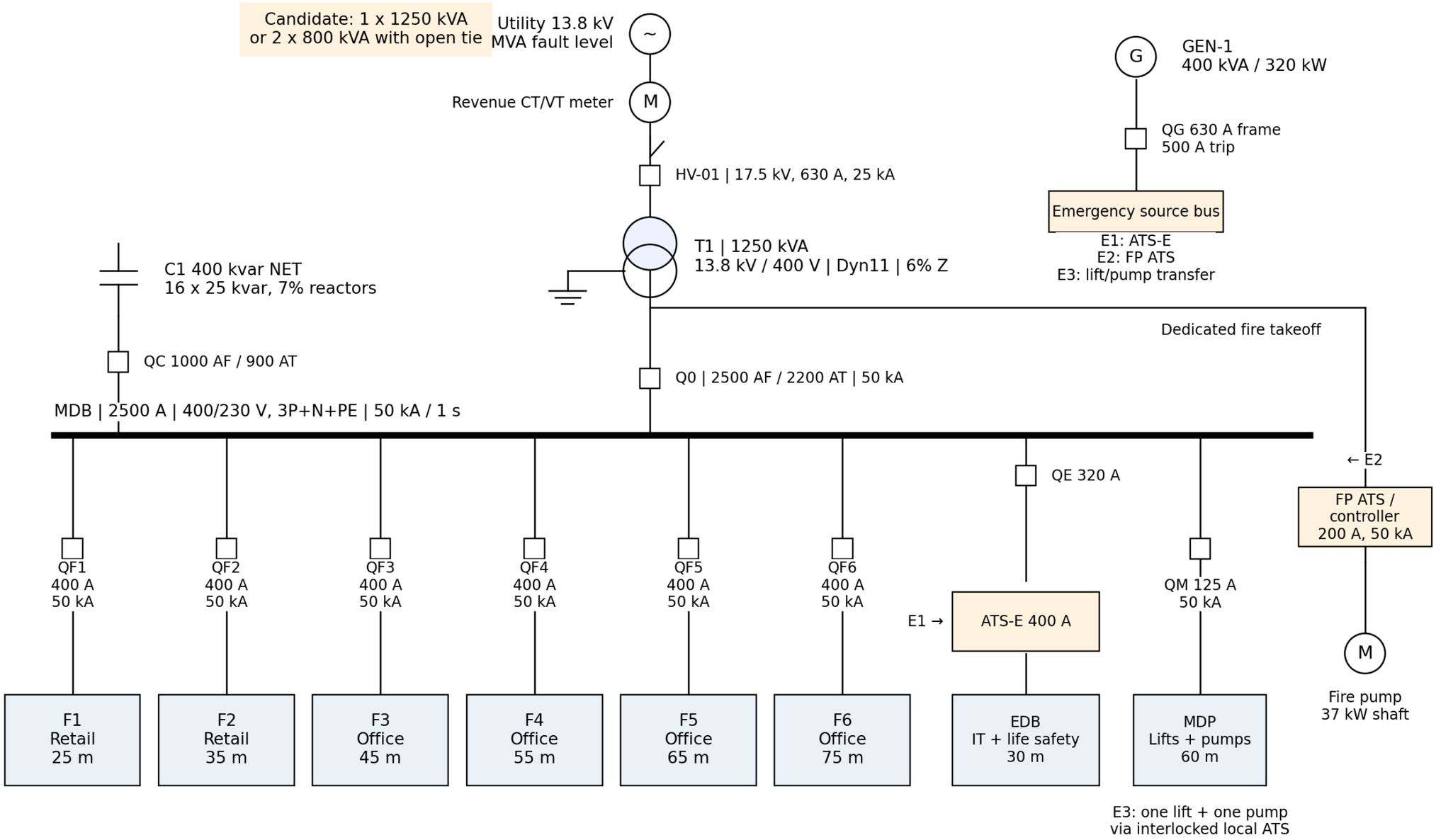
Task 2 - normal and standby paths

Table 11. Selected configuration and justification

Decision	Selected arrangement / reason
Utility service	Single 13.8 kV radial feeder; 17.5 kV class 630 A VCB/RMU with 25 kA interruption. CT/VT revenue metering on utility side of transformer.
Transformer	One 1250 kVA Dyn11 transformer. Delta primary blocks LV zero-sequence current transfer; grounded-wye secondary provides a neutral.
LV main system	400/230 V TN-S. 2500 A main board; 10 parallel sets of 300 mm² phase conductors, 200% neutral. Segregated cable tiers maintain the assumed thermal environment.
Distribution	8 normal outgoing feeders: F1-F6, EDB and MDP. One separate fire-pump takeoff ahead of main building breaker; one APFC outgoing circuit.
Main switching	2500 AF / 2200 AT ACB; utility isolation and transformer protection upstream; lockable visible HV isolation for maintenance.
Emergency supply	400 kVA generator, open-transition four-pole essential ATS and listed fire-pump ATS/controller. One lift and one duty water pump transfer locally. Normal load shedding prevents full-building generator pickup.
Grounding	Transformer neutral bonded at source; separate PE and N downstream; generator source bond coordinated with switched-neutral ATS. No additional downstream N-PE bonds.
Metering / APFC CT	PCC meter PF controls APFC using transformer-aware target; local MDB CT measurements with transformer VAR compensation or PCC transducer feedback.

A2 uses two independently protected 800 kVA transformers with split LV boards and a normally open bus tie. Break-before-make interlocks prevent parallel operation. A1 is simpler and lower cost; A2 permits reduced service during one transformer outage but still shares the single utility feeder.

Preliminary single-line diagram



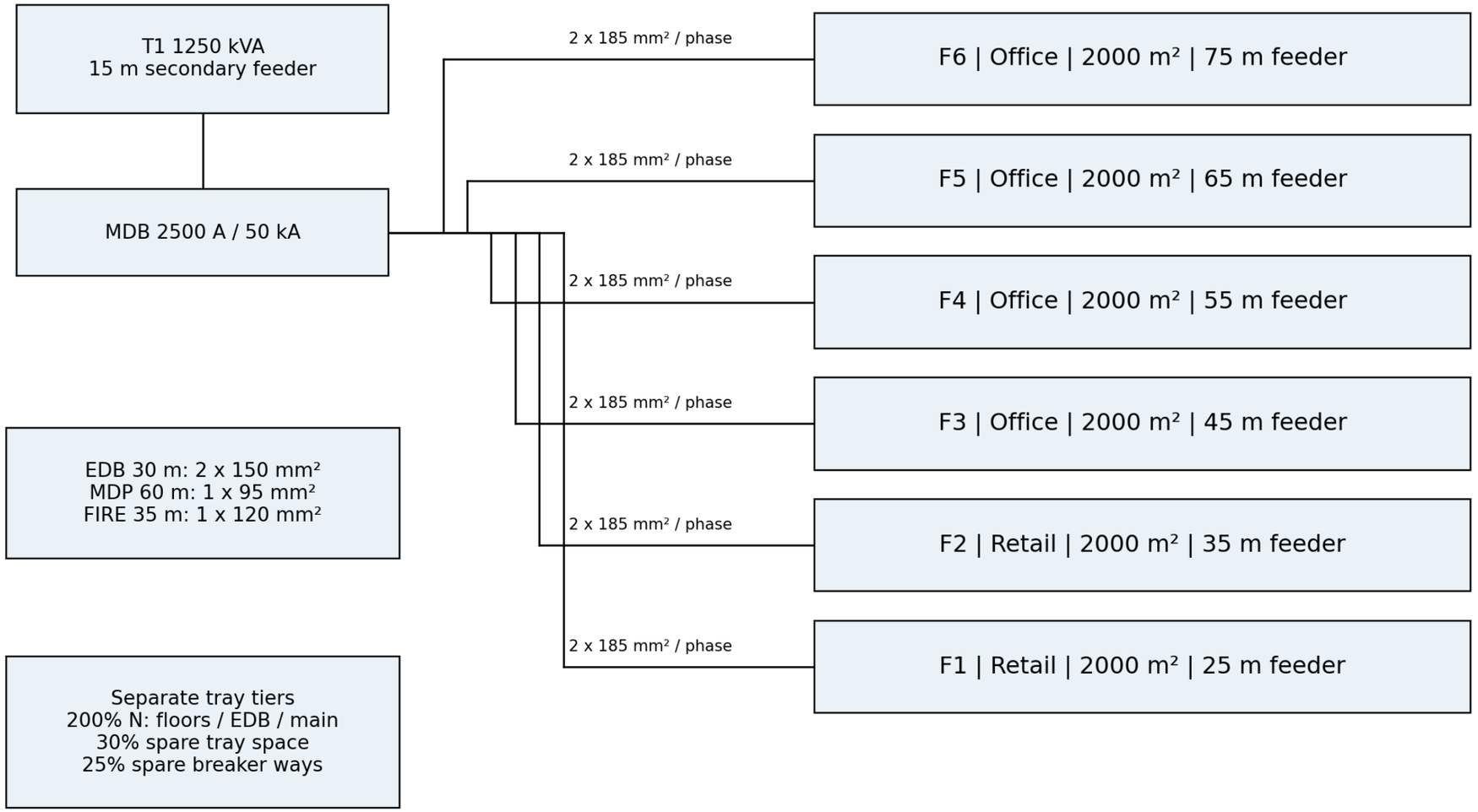
Legend: ~ utility | overlapping circles transformer | □ breaker | M meter/motor (labelled) | G generator

All LV switchgear / ATS conditional withstand: 50 kA. E1-E3 are matched emergency-source connections; no utility parallel.

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Figure 2. E-01: preliminary topology with both transformer candidates identified.

Distribution configuration | dedicated radial floor feeders



Each floor has one independent feeder. No daisy-chain floor supply.
Normal lighting, receptacles, tenant loads and VRF share each floor panel.
Fire pump bypasses the main building breaker; protected by its listed controller.

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Figure 3. E-03: individual floor feeders and fixed route lengths.

7 Equipment Sizing and Selection

Task 3 - transformer capacity

```
Sload,target,10 = 1018.7364/0.97 = 1050.2437 kVA
Srequired = Sload × (1+margin)/harmonic capacity factor
= 1050.2437 ×1.10/0.95 = 1216.0717 kVA
Next selected standard candidate = 1250 kVA
Network check = SPCC,10 ×1.10/0.95
= 1063.5595 ×1.10/0.95 = 1231.4899 kVA
Result: 1231.49 < 1250 kVA: OK
Rated HV current = 1250/(√3 ×13.8) = 52.2962 A
Rated LV current = 1250/(√3 ×0.400) = 1804.2196 A
xT = √(0.06² - 0.01²) = 0.0591608 pu
ZT = (0.01 + j0.0591608)(400²/1250000)
= 0.001280 + j0.007573 Ω per phase, LV referred
```

Table 12. Calculated transformer loading

Case	kVA at PCC	Winding-current loading	Effective thermal loading /0.95
Present corrected	788.63	61.61%	64.85%
Year 10 corrected	1,063.56	83.12%	87.49%
Year 10 low utility	1,063.67	85.26%	89.75%
Year 10 fire + low utility	1,106.14	88.67%	93.34%

Winding loading is calculated from secondary current divided by rated current; PCC kVA divided by nameplate differs because of the tap and losses. Thermal capability uses the stated 0.95 planning allowance. Transformer taps are selected to produce a 1.025 pu equivalent no-load LV source at nominal utility voltage; the actual nameplate tap ratio is verified during procurement.

Task 3 - feeder thermal method and conductor selection

```
I1 = S ×1000/(√3 VLL)
Irms = I1 ×√(1+THDi²); THDi = 0.15
Idesign = 1.25 × S10 ×1000/(√3 ×380) ×√(1+0.15²)
Iz = n × Itable ×kT ×kG ×kN
= n × Itable ×0.87 ×0.80 ×0.86 = n ×Itable ×0.59856
Acceptance: Idesign ≤ breaker long-time setting ≤ Iz
```

The 125% continuous-load multiplier is applied conservatively to all coincident normal load, not just the lighting portion. It is not asserted as a universal PEC rule for every device. Fire-pump short-circuit protection is an explicit special case. Breaker frame size and adjustable long-time pickup are distinguished throughout.

Table 13. Reference baseline and calculated derating [7]

Cu area mm²	Method-C 70°C base A	Calculated A/run
50	144	86.19
70	184	110.14
95	223	133.48
120	259	155.03
150	299	178.97
185	341	204.11
240	403	241.22
300	464	277.73

Using the conservative 70°C baseline for XLPE also avoids depending on 90°C termination capability. Factor 0.87 is the published 40°C correction for that baseline; 0.80 is the assumed physical grouping envelope; 0.86 is an additional conservative loaded-neutral allowance. Do not bundle all parallel circuits together and retain the same grouping factor.

Table 14. Calculated normal feeder inequalities at year 10, 380 V, THDi allowance

Tag	Cable per phase	Design A	Breaker A	Cable A	Result
F1	2 × 185 mm²	330.95	400	408.22	OK
F2	2 × 185 mm²	330.95	400	408.22	OK
F3	2 × 185 mm²	311.39	400	408.22	OK
F4	2 × 185 mm²	311.39	400	408.22	OK
F5	2 × 185 mm²	311.39	400	408.22	OK
F6	2 × 185 mm²	311.39	400	408.22	OK
EDB	2 × 150 mm²	201.04	320	357.94	OK
MDP	1 × 95 mm²	110.33	125	133.48	OK

Task 3 - full feeder substitutions and main circuit

```
F1: S10 = √[(112.3300×1.343916)² + (61.8462×1.343916)²]
= 172.3307 kVA; Idesign =1.25×172.3307×1000/(√3×380)×1.011187
= 330.9482 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
330.95 ≤ 400 ≤ 408.22 A: OK

F2: S10 = √[(112.3300×1.343916)² + (61.8462×1.343916)²]
= 172.3307 kVA; Idesign =1.25×172.3307×1000/(√3×380)×1.011187
= 330.9482 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
330.95 ≤ 400 ≤ 408.22 A: OK

F3: S10 = √[(106.0800×1.343916)² + (57.4837×1.343916)²]
= 162.1486 kVA; Idesign =1.25×162.1486×1000/(√3×380)×1.011187
= 311.3943 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
311.39 ≤ 400 ≤ 408.22 A: OK

F4: S10 = √[(106.0800×1.343916)² + (57.4837×1.343916)²]
= 162.1486 kVA; Idesign =1.25×162.1486×1000/(√3×380)×1.011187
= 311.3943 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
311.39 ≤ 400 ≤ 408.22 A: OK

F5: S10 = √[(106.0800×1.343916)² + (57.4837×1.343916)²]
= 162.1486 kVA; Idesign =1.25×162.1486×1000/(√3×380)×1.011187
= 311.3943 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
311.39 ≤ 400 ≤ 408.22 A: OK

F6: S10 = √[(106.0800×1.343916)² + (57.4837×1.343916)²]
= 162.1486 kVA; Idesign =1.25×162.1486×1000/(√3×380)×1.011187
= 311.3943 A
Iz = 2×341×0.87×0.80×0.86 = 408.2179 A
311.39 ≤ 400 ≤ 408.22 A: OK

EDB: S10 = √[(74.0000×1.343916)² + (24.3226×1.343916)²]
= 104.6840 kVA; Idesign =1.25×104.6840×1000/(√3×380)×1.011187
= 201.0378 A
Iz = 2×299×0.87×0.80×0.86 = 357.9389 A
201.04 ≤ 320 ≤ 357.94 A: OK

MDP: S10 = √[(35.0556×1.343916)² + (24.4689×1.343916)²]
= 57.4533 kVA; Idesign =1.25×57.4533×1000/(√3×380)×1.011187
= 110.3348 A
Iz = 1×223×0.87×0.80×0.86 = 133.4789 A
110.33 ≤ 125 ≤ 133.48 A: OK
```

Main transformer-to-board circuit

```
Conservative normal main basis: P10 / minimum true-PF design proxy
Idesign =1.25×(1018.7364/0.95)×1000/(√3×380)×√(1+.15²)
= 2059.3761 A
Select 10 parallel sets ×300 mm² Cu per phase
Iz =10×464×0.59856 = 2777.3184 A
2059.38 ≤ Ir 2200 ≤ Iz 2777.32 A: OK
Frame/bus =2500 A; main interruption=50 kA; Icw=50 kA/1 s
```

The main basis conservatively applies both a 0.95 PF proxy and an RMS harmonic allowance. A 2,500 A frame with 2,200 A long-time setting protects the cable; transformer 49/87T and HV relay protect transformer thermal capacity. Parallel cores have identical lengths, areas, material and terminations; phase sets are arranged symmetrically.

Task 3 - neutrals, PE and fire-pump special circuit

Table 15. Neutral and protective conductor schedule

Circuit	Neutral provision	PE provision
T1-MDB	20 ×300 mm² (200% of phase aggregate)	10 ×150 mm²; main bond / earth bar lead 300 mm²
F1-F6 each	4 ×185 mm² (200% of phase aggregate)	2 ×120 mm²
EDB	4 ×150 mm² (200% of phase aggregate)	2 ×95 mm²
MDP	1 ×95 mm² full-size N	1 ×95 mm², upsized by adiabatic check
Fire pump	None for 3-phase motor; control supply per listed assembly	1 ×70 mm²
Capacitor bank	No power neutral; separate fused control supply	4 ×120 mm² with 4 ×240 mm² phase sets
Generator to emergency bus	3 ×240 mm² full neutral; sensitive EDB retains 200% downstream N	3 ×120 mm²

The actual harmonic-neutral current cannot be inferred from total THDi alone. At the assumed 20% triplen envelope, a conservative in-phase third-harmonic neutral component is 3×0.20×Iphase =0.60 Iphase; 200% neutral area also accommodates phase imbalance and expansion. Neutral terminations and busbars must carry the specified aggregate current.

```
PE adiabatic screen: SPE ≥ I√t/k; kCu,XLPE =143 A√s/mm²
Common main bond: 38094.9×√0.35/143 = 157.60 mm²
Select 300 mm² common bond: OK
Feeder-origin PE worst case: 37435.5×√0.08/143 = 74.04 mm²
Select 95 mm² MDP PE (50 mm² preliminary would fail): OK
Fire motor input =37/0.90 = 41.1111 kW
Fire FLC = 41.1111×1000/(√3×400×0.85) = 69.8103 A
Fire conductor minimum =1.25×69.8103 = 87.2629 A
120 mm² Iz =259×0.59856 = 155.0270 A: OK
Assumed locked rotor =6×69.8103 = 418.8620 A
```

Fire controller/ATS is a listed 200 A assembly, 50 kA short-circuit rating, with a preliminary 600 A magnetic pickup and locked-rotor-compatible protection. Its breaker rating is intentionally not constrained below ordinary cable ampacity: special fire-pump continuity requirements apply. No ordinary ground-fault trip or general building main disconnect may interrupt the dedicated fire circuit. Procurement requires the controller coordination data, not a generic MCCB substitution.

Task 3 - standby and capacitor feeder sizing

```
Pessential,10 = (74 +18.5/0.90 +7.5/0.90)×1.343916 +37/0.90
= 179.3852 kW
Qessential,10 = [24.3228 +(20.5556+8.3333)×tan(arccos .82)]
×1.343916 +41.1111×tan(arccos .85)
= 85.2654 kVAr
Sessential,10 =√(179.3852²+85.2654²) = 198.6183 kVA
Generator kW loading = 179.3852/320×100 = 56.06%
Generator kVA loading = 198.6183/400×100 = 49.65%
Generator feeder Idesign =1.25×198.6183×1000/(√3×380)×1.011187
= 381.43 A
3×240 mm² Iz =3×403×0.59856 = 723.66 A
381.43 ≤ 500 A trip ≤ 723.66 A: OK
```

The common generator source uses ground-fault alarm without trip so its fire-pump path is not interrupted; non-fire outgoing circuits retain coordinated ground-fault tripping. QG uses a 630 A frame set to 500 A for the essential feeder. It intentionally does not provide the full generator nameplate current (577.35 A) continuously. Generator controller overcurrent/voltage/frequency protection is coordinated separately. Essential ATS rating is 400 A; fire ATS is separately rated 200 A. Two local interlocked 100 A essential motor transfers are included in the generator transfer allowance.

```
Capacitor fundamental current =400000/(√3×400) = 577.35 A
Continuous sizing basis =1.35×577.35 = 779.42 A
Voltage/harmonic current screen =1.05×1.30×577.35 = 788.08 A
Select 1000 A bank feeder; 4×240 mm² Iz = 964.88 A
```

Corrective action: a 1,000 A trip would exceed 4×240 mm² derated ampacity. Use a **1,000 A frame with 900 A long-time setting**; 788.08 A ≤900 A ≤964.88 A. Drawings and final schedule explicitly state QC 1000 AF/900 AT. Each 25 kVAr step: I=25,000/(√3×400)=36.08 A; 1.35I=48.71 A; select 63 A fuse and capacitor-duty contactor, with vendor reactor thermal protection.

Task 3 - generator starting and earth electrode screen

```
Generator start screen, fire pump first: assumed locked-rotor pf=0.30
Istart =6×69.8103 = 418.8620 A
ΔVgenerator ≈ x″d (Istart/Irated)sinφ ×100
=0.15×(418.8620/577.3503)×√(1-.30²)×100
= 10.381%
ΔVfire cable =√3×418.8620×(0.006912×.30
+0.003360×√(1-.30²)) = 3.830 V
ΔVgenerator cable = 0.586 V
Total approximate starting dip = 10.381+0.957+0.147
= 11.485% <15%: OK
Running-power pickup =41.1111/320×100 = 12.85%
Starting active power is checked separately below, not equated to running kW.

Locked-rotor constant-impedance refinement at standstill:
Zm =(230.940/418.8620)(.30+j√(1-.30²))
=0.1654054+j0.5259557 Ω
Zg =(.15×400²/400000)(1+j10)/√101
=0.0059702+j0.0597022 Ω
Zcables =0.0075708+j0.0040000 Ω
Istart,actual =230.940/(Zg+Zcables+Zm)
=108.83310-358.62287j A; |I|=374.7733 A
Vmotor =√3 |I×Zm|=357.8967 V; dip=10.5258% <15%
Pstart,source =Re[3×230.940×conj(I)]/1000=75.40178 kW
Starting active-power step =75.40178/320×100=23.5631% <25%: OK
```

This is a subtransient screening calculation, not a dynamic motor acceleration simulation. Fire-pump priority drops optional lift/pump loads and sequences IT/UPS pickup or recharge. The engine step-response curve and pump acceleration torque must confirm the screen. Emergency voltage recovery to 0.95-1.05 pu is required after starting. No APFC is connected in islanded generator operation.

```
Rod resistance ≈ρ/(2πL)[ln(4L/d)-1]
=100/(2π×3)[ln(4×3/0.016)-1] = 29.815 Ω
Assume 12 spaced rods, aggregate efficiency 0.65:
Re ≈ 29.815/(12×0.65) = 3.822 Ω <5 Ω: OK (estimate)
```

Use a bonded perimeter ring, rod spacing at least one rod length where practicable, structural bonding and accessible test links. Measure soil/electrode resistance and touch/step performance. The 5 Ω project target is not a substitute for automatic disconnection through metallic PE conductors; no site-specific earthing safety certification is implied.

8 Voltage-Drop / Voltage-Profile Analysis

Task 4 - hand method

```
ΔVLL =√3 I (R cosφ + X sinφ)L
Rhot =23.7/A Ω/km; X60Hz =0.08×60/50 =0.096 Ω/km
For n equal parallel sets: Zroute =(R+jX)L/n
Equivalent form: ΔVLL =(P W ×Rroute +Q var ×Xroute)/VLL
Transformer regulation ≈ loadingpu ×(rpu cosφ +xpu sinφ)×100
Total wiring drop = main cable + floor feeder + branch reserve
Load voltage also includes source and transformer regulation.
```

Cable R is a conservative hot design estimate; X is the 60 Hz screening value from the manufacturer guide [10,11]. These are not exact installed impedances. The detailed load flow uses constant-power loads and voltage-dependent capacitors, unlike the single-pass 400 V hand formula.

Farthest feeder F6 - explicit substitution

```
P10=106.0800×1.343916=142.5626 kW
Q10=57.4837×1.343916=77.2533 kVAr
I=162.1486×1000/(√3×400)=234.0414 A; cosφ=0.879210
R=23.7/185=0.128108 Ω/km; X=.096 Ω/km
L=.075 km; parallel n=2
ΔV=√3×234.0414×[(0.128108/2)×0.879210
+(.096/2)×0.476435]×.075
=2.4075 V = 0.6019% <3%: OK

Main cable year 10 corrected ΔV = 0.3710 V
Total wiring =(0.3710+2.4075)/400×100 +1% reserve
=1.6946% <5%: OK
```

Task 4 - bus voltages, regulation and correction

Table 16. Independent solver voltages - nominal source

Bus	Present no APFC V	Present corrected V	Year 10 corrected V	Year 10 low source V
TA	398.84	404.13	402.14	391.77
MDB-A	398.46	403.86	401.77	391.39
F1	397.82	403.23	400.92	390.52
F2	397.57	402.97	400.57	390.17
F3	397.38	402.79	400.32	389.91
F4	397.14	402.55	400.00	389.58
F5	396.90	402.31	399.68	389.25
F6	396.66	402.07	399.36	388.92
EDB	397.93	403.33	401.06	390.67
MDP	396.79	402.20	399.53	389.09
FIRE	398.84	404.13	402.14	391.77

```
Year-10 transformer load angle: cosφ ≈ 0.97369
Regulation ≈ 0.831192×(.01×0.97369
+.0591608×√(1-0.97369²))×100 = 1.9298%
Low-source farthest panel = 388.916 V
After 1% branch reserve: 388.916-4
= 384.916 V >380 V: OK
```

An initial no-tap/no-APFC trial gives a year-10 minimum panel voltage of 381.20 V. Subtracting the 4 V branch reserve gives 377.20 V, below 380 V: **NOT OK**. The selected 1.025 equivalent tap and APFC correct it to 395.36 V at the load under nominal supply. No load input was changed.

At the high-source night case (20% load), maximum energized panel voltage is 419.24 V. Capacitors are stepped down automatically. At strictly zero load, 1.025×1.025×400=420.25 V; this is 0.25 V above the planning load limit and requires checking the actual tap ratio/utility high-voltage setting at commissioning. No-load apparatus insulation has ample rating, but a ±5% guarantee across every possible utility condition cannot be inferred from one fixed tap.

Task 4 - feeder voltage drop and loading register

Table 17. Calculated cable routes and fundamental loading

Tag	m	R Ω	X Ω	Drop Y0 %	Drop Y10 %	Load Y0 %	Load Y10 %
F1	25	0.001601	0.001200	0.159	0.213	44.98	60.79
F2	35	0.002242	0.001680	0.222	0.299	45.01	60.85
F3	45	0.002882	0.002160	0.269	0.361	42.37	57.29
F4	55	0.003523	0.002640	0.328	0.441	42.39	57.33
F5	65	0.004164	0.003120	0.388	0.522	42.42	57.38
F6	75	0.004804	0.003600	0.448	0.602	42.44	57.42
EDB	30	0.002370	0.001440	0.132	0.177	31.15	42.10
MDP	60	0.014968	0.005760	0.416	0.559	45.98	62.20

All normal feeders meet 3% at future load. Loading percentages use actual solved current divided by derated cable ampacity. Ampacity selections separately include 125%, minimum voltage and RMS harmonic allowances. Central APFC reduces main current; it does not remove the reactive current inside any individual floor feeder.

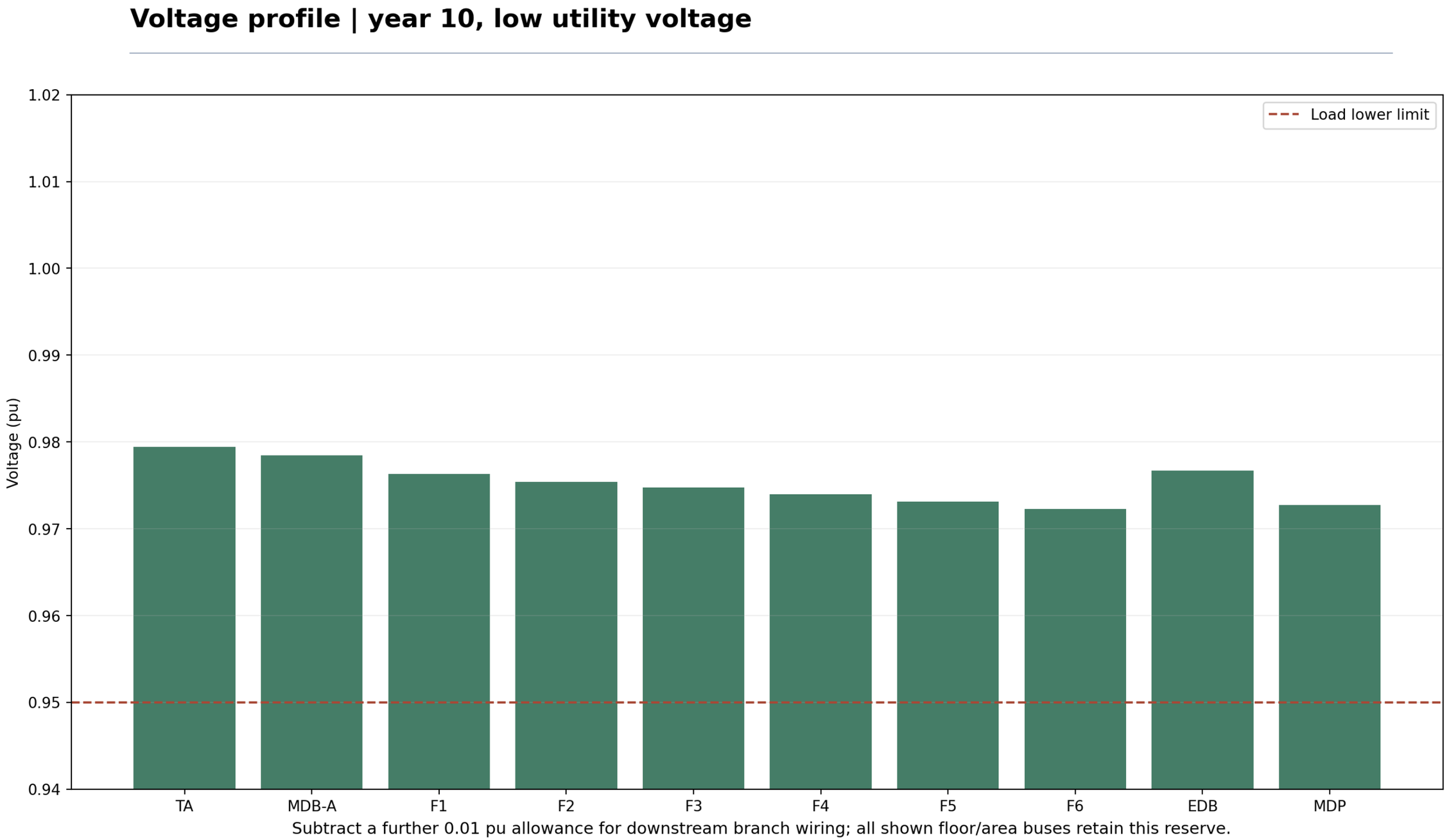


Figure 4. E-10: year-10 low-utility voltage profile, corrected design.

9 Short-Circuit Analysis

Task 5 - assumptions and equivalent impedances

Maximum three-phase bolted faults set interrupting and withstand duties. All impedances are per phase and referred to the 400 V base. Generator/utility and A2 transformers are never paralleled. The motor term is an intentionally conservative arithmetic bound using all installed future motor capacity; converter-fed motors are not claimed to provide this exact contribution. This is an IEC-style engineering screening study, not a complete sequence-network or dynamic IEC certification.

$$I''_Q = 250/(\sqrt{3}\times 13.8) = 10.459244\ \text{kA}$$

$$|Z_{Q,HV}| = c\times U^2/S''_Q = 1.10\times 13.8^2/250 = 0.837936\ \Omega$$

$$|Z_{Q,LV}| = 1.10\times 400^2/(250\times 10^6) = 0.0007040\ \Omega$$

$$R_Q = |Z_Q|/\sqrt{1+10^2} = 0.000070051\ \Omega$$

$$X_Q = 10R_Q = 0.000700506\ \Omega$$

$$Z_{HV\text{cable},LV} = (.494+j.120)\times .020\times (400/13800)^2$$

$$= 0.000008301+j0.000002016\ \Omega$$

$$Z_{T,nom} = .001280+j.007573\ \Omega$$

$$K_T = .95c_{max}/(1+.6xT) = .95\times 1.10/(1+.6\times .0591608)$$

$$= 1.0091777$$

$$Z_{T,max\text{-}fault} = K_T\times .90\times Z_{T,nom}$$

$$= 0.001162573+j0.006877873\ \Omega$$

The specified 250 MVA is interpreted as maximum initial short-circuit power at PCC; c therefore appears in the source impedance as well as the equivalent voltage. This avoids silently multiplying the stated utility fault level by c a second time. The hand load flow uses the nominal source impedance without the maximum-fault voltage factor.

Task 5 - cable and motor calculations

```
Main cold cable Z =[(17.241/300)+j.096]×.015/10
= 0.000086205+j0.000144000 Ω
F6 cold cable Z =[(17.241/185)+j.096]×.075/2
= 0.003494797+j0.003600000 Ω
```

Table 18. Calculated motor bounding contribution

Motor group	Connected input kW	Connected kVA	Future kVA
HVAC	432.00	508.24	683.03
LIFT	41.11	50.14	67.38
WP	16.67	20.33	27.32
FP	41.11	48.37	65.00
ESC	16.67	20.33	27.32

```
Smotor,10 = 870.034282 kVA
Imotor,bound =4×Smotor/(√3×0.400)
=4×870.034282/(√3×0.400)
= 5023.145 A = 5.023145 kA
Iutility,bus =c×400/(√3×|ΣZ|)/1000 kA
Idesign,bus =Iutility,bus + Imotor,bound (arithmetic upper bound)
```

The same full motor bound is applied at each LV bus without attenuation. It therefore overstates many remote duties and is unsuitable for claiming exact motor contribution or selectivity. At HV it is scaled by 0.4/13.8. For A2 the same whole-building bound is retained as a conservative equipment screen even with the tie open.

```
T: ZΣ =0.001240924+j0.007580396 Ω
|ZΣ| =√(0.001240924²+0.007580396²)
=0.007681295 Ω
Iutility =1.10×400/(√3×0.007681295)/1000
=33.071783 kA
Idesign =33.071783+5.023145=38.094928 kA

MDB: ZΣ =0.001327129+j0.007724396 Ω
|ZΣ| =√(0.001327129²+0.007724396²)
=0.007837574 Ω
Iutility =1.10×400/(√3×0.007837574)/1000
=32.412342 kA
Idesign =32.412342+5.023145=37.435487 kA

F6: ZΣ =0.004821926+j0.011324396 Ω
|ZΣ| =√(0.004821926²+0.011324396²)
=0.012308246 Ω
Iutility =1.10×400/(√3×0.012308246)/1000
=20.639344 kA
Idesign =20.639344+5.023145=25.662489 kA
```

Task 5 - interrupting and withstand requirements

Table 19. Calculated maximum three-phase duty and selected equipment

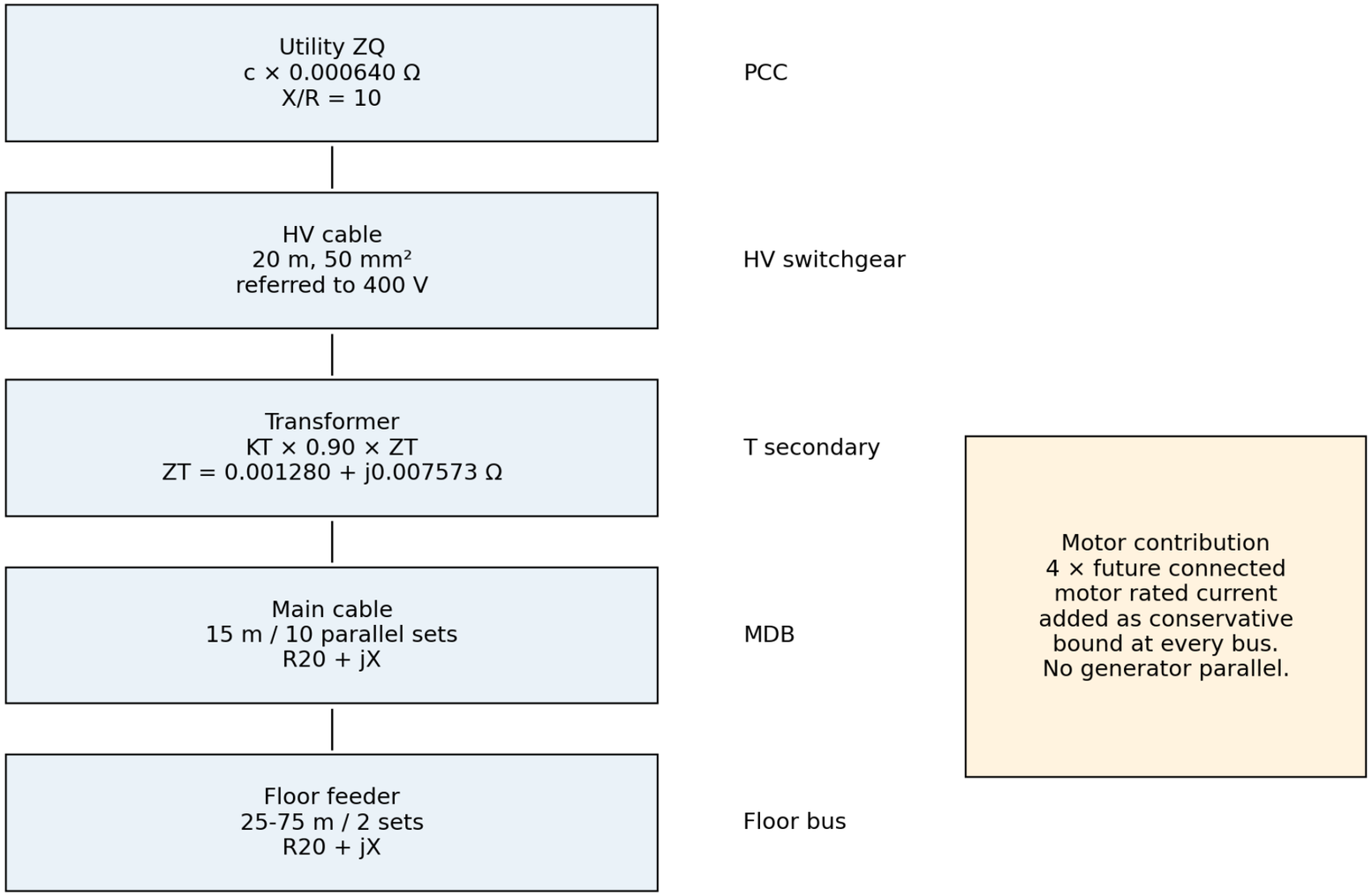
Bus	kV	Utility kA	Motor bound kA	Total kA	Rating kA	Adequate
PCC	13.8	10.46	0.15	10.60	25	YES
HV	13.8	10.42	0.15	10.56	25	YES
T	0.4	33.07	5.02	38.09	50	YES
MDB	0.4	32.41	5.02	37.44	50	YES
F1	0.4	27.42	5.02	32.44	50	YES
F2	0.4	25.77	5.02	30.79	50	YES
F3	0.4	24.28	5.02	29.31	50	YES
F4	0.4	22.95	5.02	27.97	50	YES
F5	0.4	21.74	5.02	26.76	50	YES
F6	0.4	20.64	5.02	25.66	50	YES
EDB	0.4	26.30	5.02	31.32	50	YES
MDP	0.4	13.96	5.02	18.98	50	YES
FIRE	0.4	20.15	5.02	25.17	50	YES

```
κ =1.02 +.98 exp(-3R/X)
ip,bound =κ√2 Iutility +2√2 Imotor,bound
Maximum calculated LV making-current bound = 89.962 kA peak
Selected LV Icu=Ics=50 kA; Icw=50 kA/1 s; Ipk≥105 kA
Maximum rms duty =38.095 kA <50 kA: OK
Peak duty =89.962 kA <105 kA: OK
```

All downstream 400 V protective assemblies are specified for 50 kA at their actual operating voltage, avoiding reliance on unverified cascading tables. ATS ratings are conditional short-circuit withstand ratings with the specified upstream protective device, not an interrupting claim. HV equipment is 17.5 kV / 25 kA, with compatible making and short-time ratings.

```
Generator initial symmetrical terminal fault screen
I″g ≈ Irated/x″d =577.3503/.15 = 3849.00 A
Generator I″g ≈ 3.849 kA; utility isolated
Minimum utility-fed 3-phase remote value (hot cables, c=.95)
= 10.104 kA; feeder short-time pickup is well below this.
```

Generator fault current decays; its initial subtransient value alone cannot prove delayed breaker operation. Generator electronic protection and the manufacturer decrement curve must confirm island-mode clearance. Earth faults require the neutral/PE return and actual zero-sequence data; a minimum-loop sensitivity check is provided in Section 10.



I″k = c Un / (√3 |ΣZ|). Independent check: invert the nodal admittance matrix and use Zbus,kk.

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Prepared by John Ray O. Dalida | EEA108D-E06 | Group 1 | Mapua University

Figure 5. E-07: per-phase impedance diagram used for hand and independent fault calculations.

10 Protection Design

Task 6 - protection philosophy and settings

The scheme isolates floor/area faults before the building main where the device curves permit. Transformer differential, thermal and restricted earth-fault protection cover internal and low-level faults. Settings below are engineering starting points, not a certified selective-coordination result. Curve overlays and manufacturer let-through data are required before energization [3,4].

Table 20. Preliminary coordination and setting table

Device / protected zone	Long-time / thermal	Short / instantaneous	Ground / delay
HV-01 / T1 intake	51 primary pickup 65 A; 0.6 s definite backup	50 primary 800 A, 0.05 s; 87T differential	51N 10 A / 0.6 s; REF sensitive transformer zone
T1 internal	49 thermal alarm 90%; trip at capability curve; 87T 0.3 pu, slope 30% assumed	2nd-harmonic inrush restraint 15% assumed; 87T fast trip	REF and winding-temperature trip; sensor/control DC supply
Q0 / main LV cable-MDB	2500 AF, Ir=2200 A; long-time 12 s at 6Ir	Isd=6Ir=13.2 kA, 0.30 s; I OFF; ZSI	Ig=440 A, 0.30 s; zone restraint from floor feeders
QF1-QF6 / floor feeders	400 AF/AT; long-time 8 s at 6Ir	Isd=2.4 kA /0.08 s; Ii=4.0 kA	Ig=80 A /0.10 s
QE / EDB normal feeder	400 AF /320 AT; 8 s at 6Ir	Isd=1.92 kA /0.08 s; Ii=3.2 kA	Ig=64 A /0.10 s; life-safety downstream coordination
QM / MDP	160 AF /125 AT; motor starts checked	Isd=.75 kA /0.08 s; Ii=1.25 kA	Ig=25 A /0.10 s
QC / APFC feeder	1000 AF /900 AT; step fuses 63 A	Capacitor-duty switching; Ii per inrush study	Reactor thermal trip; discharge, overvoltage and harmonic alarm
QG / generator feeder	630 AF /500 AT; generator controller 51V	Isd=1.5 kA /0.10 s; instantaneous per decrement curve	Common generator GF alarm only; GF tripping confined to non-fire outgoing feeders
Fire controller	200 A listed assembly; no ordinary overload shutdown	600 A magnetic screen >419 A assumed LRA; locked-rotor protection per listing	Alarm earth faults; no ordinary GF trip in fire supply
A2 incomers / tie	1600 AF /1400 AT each; tie 1600 AF /1400 AT	Isd=8.4 kA /0.30 s; ZSI; tie N.O.	280 A /0.30 s; 3-breaker interlock

Task 6 - inrush, backup and clearing checks

```
T1 rated HV current =1250/(√3×13.8)=52.296 A
Assumed inrush peak-equivalent envelope =12×52.2962=627.555 A
HV high-set 800 A >627.55 A: inrush screen OK
LV main short-time pickup referred to primary
=13200×400/13800=382.609 A
HV 51 backup delay .60 s - Q0 .30 s =.30 s grading margin
Q0 .30 s - QF .08 s =.22 s nominal grading margin
```

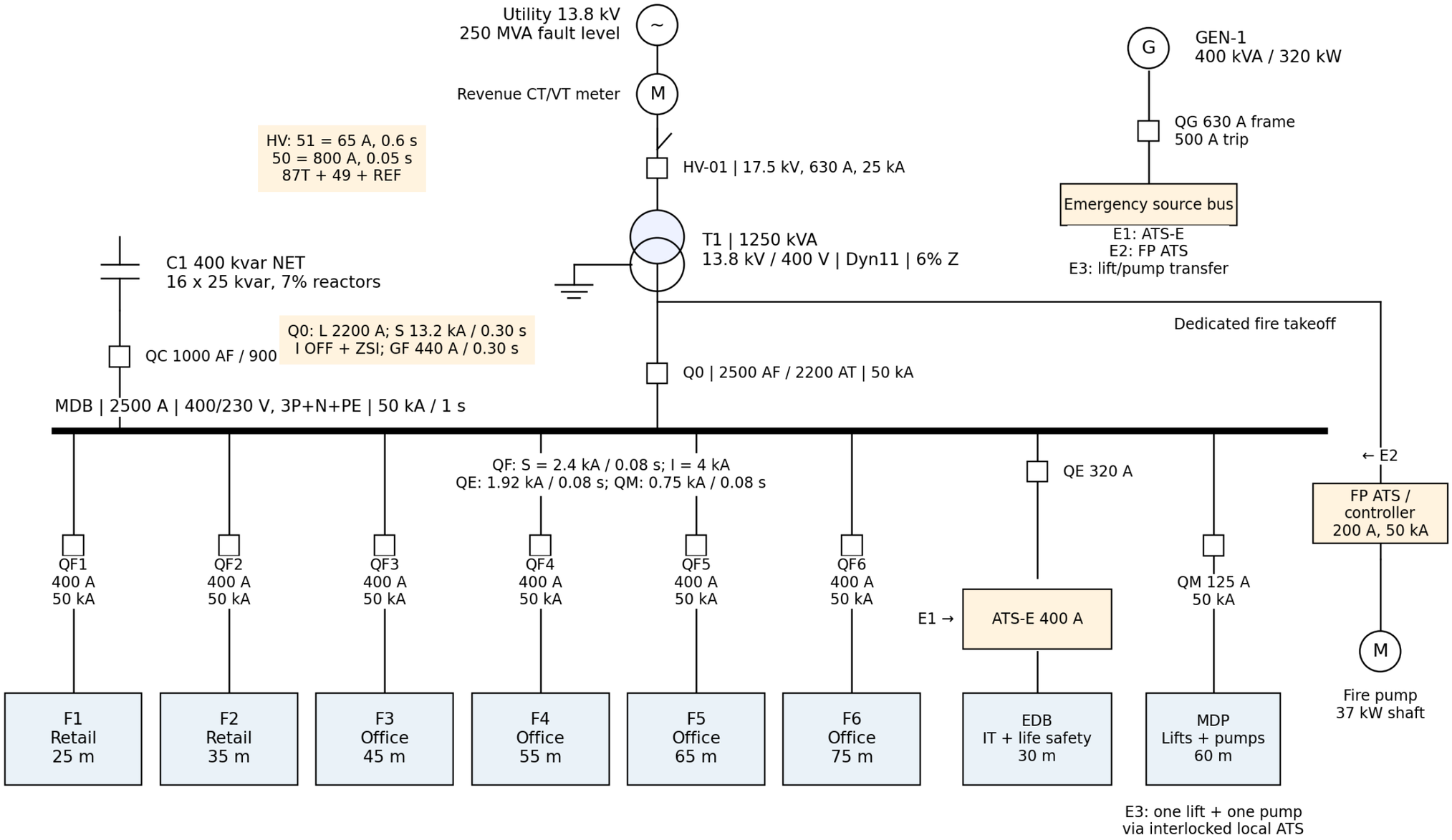
Instantaneous overlap can remove this simple time margin at high fault current. ZSI permits fast main clearing when no downstream restraint arrives; with restraint, the main waits for the feeder. Specify IEC 60947-2 category-B main gear with adequate short-time withstand. Do not infer total selectivity from this table alone. HV cable/transformer protection must also remain below transformer through-fault damage curves.

```
F6 TN-S loop sensitivity screen (assume Z0 transformer≈Z1):
Zloop ≈ZT +2Zsource +2Zmain +Zphase,F6 +ZPE,F6
=0.013855+j0.016335 Ω
If,earth,min≈.95×230.940/|Zloop|=10242.9 A
10242.9 A >80 A feeder GF pickup: sensitivity screen OK
```

This return-loop approximation is deliberately identified as a screen. Actual transformer zero-sequence impedance, bonding and generator neutral switching are needed for a formal line-to-earth study. Solid neutral continuity is monitored; four-pole transfer prevents parallel neutral-earth paths between utility and generator sources.

Fire-pump protection is coordinated independently from discretionary loads. The dedicated connection is ahead of Q0, the fire ATS has both normal and standby sources, and its disconnect/controller is lockable in the normal operating position. Utility-side common faults can still interrupt normal power; generator provides the alternate source.

Protection arrangement | preliminary settings



Legend: ~ utility | overlapping circles transformer | □ breaker | M meter/motor (labelled) | G generator

All LV switchgear / ATS conditional withstand: 50 kA. E1-E3 are matched emergency-source connections; no utility parallel.

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Figure 6. E-04: annotated final topology; settings are preliminary and require device curves.

11 Future Expansion Analysis

Task 7 - ten-year verification

Table 21. Calculated growth, selected APFC steps and final transformer loading

Year	Demand kW	Switched kVAr	PCC PF	T1 load %	Min panel pu
0	758.04	250	0.9743	61.61	1.0052
1	780.78	250	0.9718	63.64	1.0041
2	804.20	275	0.9764	65.26	1.0044
3	828.33	275	0.9738	67.41	1.0033
4	853.18	275	0.9712	69.65	1.0021
5	878.77	300	0.9752	71.46	1.0023
6	905.13	300	0.9726	73.83	1.0010
7	932.29	325	0.9761	75.80	1.0011
8	960.26	325	0.9734	78.31	0.9998
9	989.06	325	0.9706	80.92	0.9984
10	1,018.74	350	0.9737	83.12	0.9984

The 400 kVAr enclosure is fitted from the start; the controller progressively uses more steps as demand grows. The year-10 low-utility case needs 375 kVAr; the simultaneous fire/low-utility case uses the available bank automatically. Standard operation stays below the transformer harmonic-adjusted capability, and normal feeder inequalities already use future load at 380 V.

```
Approximate growth trigger preserving 10% reserve and 0.95 capability:
Pcapacity≈1250×.95×.97/1.10=1047.159 kW
n≈ln(1047.159/758.036)/ln(1.03)=10.93 years
Network losses make this a screening estimate; repeat the model before year 10.
```

Table 22. Upgrade / operating action register

Item	Year / trigger	Required action
Transformer	No planned replacement in years 0-10; review around year 10-11	Re-run actual demand and harmonic loss study; use reserved second-transformer space if margin is exhausted.
Normal feeders	No upgrade in years 0-10 under frozen occupancy	Recheck with tenant-specific connected loads before permitting fit-out increases.
APFC	Automatic stages added in service according to growth table	Keep all 16 steps available; alarm stage failures; add capacity only after harmonic/resonance review.
APFC failure	Any year; especially year-10 fire/low-utility	Shed discretionary HVAC/tenant load or restore APFC; uncorrected future duty can exceed harmonic-adjusted thermal allowance.
A2 transformer outage	Any year when full load exceeds one 800 kVA unit	Shed 45% normal demand before closing tie; generator protects essential service.
Budget	Initial procurement	Current A1 estimate exceeds soft target; authorize revised budget or value-engineer with verified route/vendor data.

Provide 25% spare breaker ways, 30% spare tray/conduit capacity, reserved second-transformer footprint, accessible HV feeder provision, and a main board configured for future sectionalization. Spare bus rating alone does not authorize additional transformer fault current; any second transformer requires a new protection and fault study.

12 Facility-Specific Analysis

Task 8 - power factor correction and commercial loads

```
pf1 = 758.035556/√(758.035556²+402.418653²) = 0.8832546
φ1 = arccos(0.8832546) = 27.96251°
Q1 =P tanφ1 = 758.035556×0.5308704 = 402.418653 kVAr
φ2 = arccos(.97) = 14.06987°
Q2 = 758.035556×tan(14.06987°) = 189.981618 kVAr
Qc =P(tanφ1-tanφ2)
=758.035556×(0.5308704-0.2506236)
=212.437035 kVAr
Qc,10 load-only =212.437035×1.343916379=285.497611 kVAr
```

Load-only sizing is insufficient at an HV metering point because the transformer consumes VARs. The solver regulates the PCC displacement PF by choosing 25 kVAr steps, including transformer/cable losses and the V² output of capacitors. Install 400 kVAr **net fundamental output at 400 V**, 16×25 kVAr steps. The initial nominal-source command is 250 kVAr and the year-10 command is 350 kVAr; remaining steps accommodate low voltage and fire duty.

```
7% detuned branch: fr=f1/√p=60/√.07=226.7787 Hz
Capacitor element voltage at 400 V bus =400/(1-.07)=430.1075 V
At +5% bus voltage =420/.93=451.6129 V <480 V: OK
Required capacitor element nameplate at 480 V for 400 kvar net
Qelement,480 =400×(1-.07)×(480/400)²=535.6800 kVAr
Per 25 kvar net delta step: Cbranch =Qnet(1-p)/(3ωVLL²)
=25000×.93/(3×2π×60×400²)=128.4845 μF
Lbranch=p/(ω²C)=3.8334 mH
```

The element/reactor calculation is an ideal delta-branch equivalent; an OEM may use a different internal arrangement with the same net output. Do not purchase a 400 kVAr-at-480 V capacitor assembly and assume it supplies 400 kVAr net at 400 V with a reactor. Specify net bank output, voltage, frequency and reactor percentage together [14].

Task 8 - resonance and harmonic performance

```
Ssc,capacitor bus =U²/|Zsource+ZT+Zmain|
=.400²/0.00848383=18.85940 MVA
Bare-bank parallel resonance order hr =√(Ssc,kVA/Qc,kVAr)
At full bank hr=√(18859.403/400)=6.8665
At 350 kvar hr=√(18859.403/350)=7.3406
```

Table 23. Bare-capacitor resonance screen - not a detuned-bank frequency scan

Switched kVAr	Ssc MVA	hr	3rd / 5th / 7th vicinity
25	18.859	27.466	Separated at this step
50	18.859	19.421	Separated at this step
100	18.859	13.733	Separated at this step
150	18.859	11.213	Separated at this step
200	18.859	9.711	Separated at this step
250	18.859	8.685	Separated at this step
300	18.859	7.929	Separated at this step
350	18.859	7.341	Review
400	18.859	6.866	Review

Some switching states approach the 7th harmonic; plain capacitors are therefore **NOT OK**. The 7% series reactor moves the LC series tuning to order 3.78, below the 5th and away from 3, 5 and 7. The simple hr formula applies to a bare shunt bank, not the complete detuned network. A measured harmonic spectrum and multi-step frequency scan are still needed to confirm that no new amplification occurs. Third-harmonic neutral currents require neutral design and load mitigation; a delta detuned bank does not solve them.

```
At THDi=15%, true PF ≈ displacement PF/√(1+THDi²)
Present true PF ≈0.974317/√(1+.15²)=0.963538
Future true PF ≈0.973694/√(1+.15²)=0.962921
Both >.95 under the assumed spectrum envelope: OK
```

A central bank is selected for simple maintenance and low switching count. It releases transformer and main-feeder capacity but does not materially reduce floor cable losses. Distributed banks could reduce riser losses but introduce many contactors, controllers and resonance locations. Install PCC feedback, anti-hunting delay, step rotation, discharge interlocks and leading-PF blocking. Block APFC on generator supply.

Task 8 - quantified benefits and honest payback

Table 24. Calculated before/after correction at nominal utility

Year	Metric	Before	After
0	PCC kVA	885.579	788.631
0	Main current A	1,247.203	1,109.081
0	T1 loading %	69.255	61.611
0	PCC displacement PF	0.868	0.974
0	Main cable ΔV V	0.372	0.277
0	T1 + main I²R loss kW	6.548	5.182
10	PCC kVA	1,202.920	1,063.559
10	Main current A	1,696.212	1,497.112
10	T1 loading %	94.142	83.119
10	PCC displacement PF	0.862	0.974
10	Main cable ΔV V	0.506	0.370
10	T1 + main I²R loss kW	12.101	9.433

```
Present upstream I²R saving =6.548221-5.182236
=1.365985 kW
Egross =1.365985×8760×.439375=5257.575 kWh/year
Ebank =(250×.005)×8760×.5875=6433.125 kWh/year
Ebank feeder =3I²R×8760×.439375=189.257 kWh/year
Enet =5257.575-6433.125-189.257
=-1364.808 kWh/year
Net present annual energy value =-1364.808×12
=PHP -16,377.69/year
Avoided PF penalty =PHP 0/year (no verified tariff penalty assumed)
Installed bank cost =950000×1.15×1.15=PHP 1,256,375.00
```

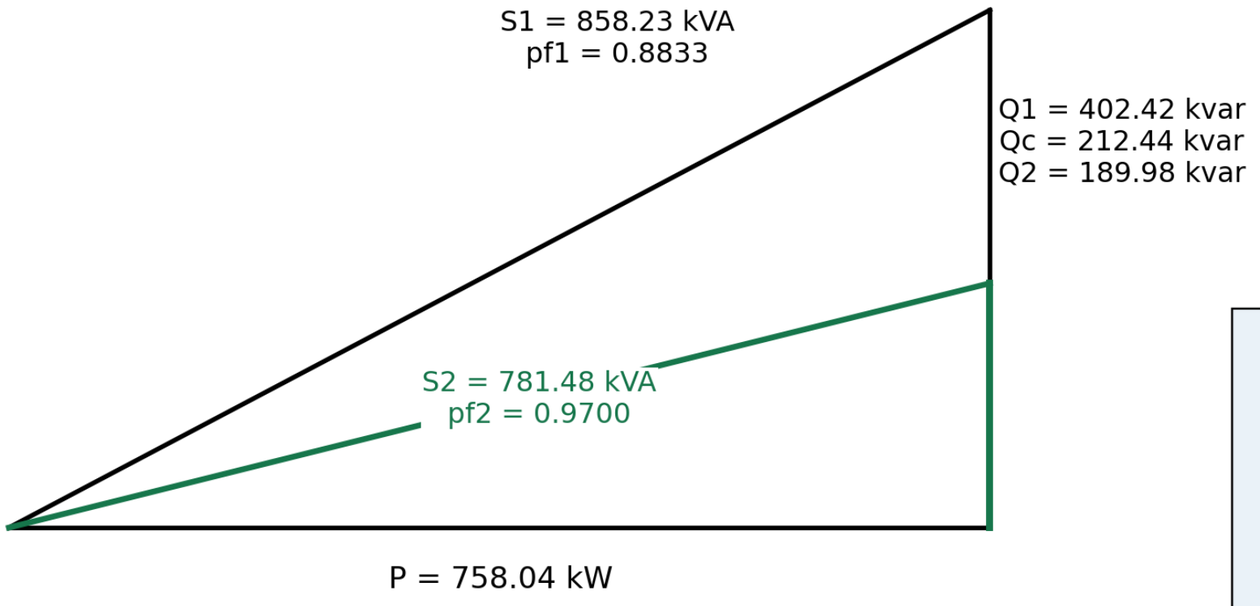
Payback result: no positive energy-only payback at present load; the assumed detuned-bank losses exceed the upstream I²R saving. This is a capacity, voltage and PF-compliance investment, not a justified electricity-bill savings claim.

```
Year-10 net saving =897.407 kWh/year×PHP12
=PHP 10,768.88/year
Year-10-only simple payback =1256375.00/10768.88
=116.67 years
Released PCC capacity now =96.948 kVA; year10=139.361 kVA
```

The annual loss estimate scales peak upstream losses with the explicit loss factor and bank losses with load factor; it is an approximation, not a time-series simulation. Utility billing may include different demand and PF provisions. No discount, demand-charge saving or penalty avoidance is booked without an applicable contract.

Power factor correction | load triangles and supply benefits

Load-only calculation: 212.44 kvar required initially.
Installed bank: 400 kvar NET at 400 V, in 16 automatic 25 kvar steps.
Solver includes transformer VARs and V² dependence: 250 kvar switched initially; 350 kvar at year 10.



Central APFC reduces transformer and main feeder current.
Floor feeder load VARs remain downstream.

7% detuning at 60 Hz: fr = 60 / √0.07 = 226.8 Hz. Do not substitute a 50 Hz reactor rating.

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Figure 7. E-06: load-only power triangles and transformer-aware APFC selection.

13 Design Alternatives

Alternative 1 - single 1,250 kVA transformer

A1 is the complete design in Sections 7-12. Normal supply is radial; essential service has the standby generator. It has no transformer N-1 capability. If T1 is unavailable, normal floors are de-energized and only the defined essential loads continue on generator.

Alternative 2 - two 800 kVA transformers with open tie

Section A supplies F1/F3/F5 and MDP; section B supplies F2/F4/F6 and EDB. The dedicated fire takeoff is on transformer A. Each transformer has its own HV breaker/relay, 1600 A LV section, 1600 AF /1400 AT incomer and 6 parallel 300 mm² main sets. Each bus has a 200 kVAr net detuned bank with 16×12.5 kVAr steps; aggregate switching is represented in 25 kVAr increments. Full transformer paralleling is prohibited.

```
A2 section A: P0=112.3300+106.0800+106.0800+35.0556
= 359.5456 kW; Q0=201.2825 kVAr
S10,target≈359.5456×1.343916/.97=498.1435 kVA
Srequired with margin/capability=498.1435×1.10/.95
= 576.7977 kVA <800 kVA: OK

A2 section B: P0=112.3300+106.0800+106.0800+74.0000
= 398.4900 kW; Q0=201.1362 kVAr
S10,target≈398.4900×1.343916/.97=552.1002 kVA
Srequired with margin/capability=552.1002×1.10/.95
= 639.2740 kVA <800 kVA: OK
```

Although 630 kVA is also a standard size, the 800 kVA candidate is retained to provide better reduced-service N-1 capability and expansion room. It is an explicitly evaluated option rather than a claim that 800 kVA is the immediate next size after the normal-load arithmetic.

Alternative 2 - capacity, voltage, faults and protection

A2 section A: Idesign=1.25×(483.1992/.95)×1000/(√3×380)×1.011187
=976.7869 A ≤1400 A ≤6×464×.59856=1666.3910 A: OK

A2 section B: Idesign=1.25×(535.5372/.95)×1000/(√3×380)×1.011187
=1082.5883 A ≤1400 A ≤6×464×.59856=1666.3910 A: OK

ZT,A2 =(0.011+j√(.06²-.011²))×400²/800000
=0.0022000+j0.0117966 Ω
KT,A2=1.0092817; apply .90 impedance tolerance
Main A2 Zcold=[17.241/300+j.096]×.015/6
=0.00014367+j0.00024000 Ω
At T bus: Iutility=1.10×400/(√3×|0.0020767+j0.0114180|)/1000
=21.8894 kA; total bound=26.9126 kA <50 kA: OK
MDB bound=26.4289 kA; F6 bound=20.6145 kA

Table 25. A2 independent-solver capacity and voltage

Case	TA load %	TB load %	Minimum panel V	PCC PF
Present	45.65	50.42	403.09	0.9764
Year 10	61.85	68.30	400.31	0.9711
Year 10 low utility	63.40	70.04	389.90	0.9722
Year 10 fire	67.32	67.95	400.19	0.9725

A2 floor/area cable selections are identical to A1, hence each satisfies the same future-load ampacity and <3% feeder-drop calculations. Voltage improves because each transformer supplies roughly half the building. Use the same 50 kA floor assemblies; fault requirements are lower but a common specification simplifies maintenance. Individual HV 51 pickups are 42 A (800/(√3×13.8)=33.47 A rated); high-set 600 A is above the assumed 12×33.47=401.6 A inrush screen. Differential and thermal protection are provided for each unit.

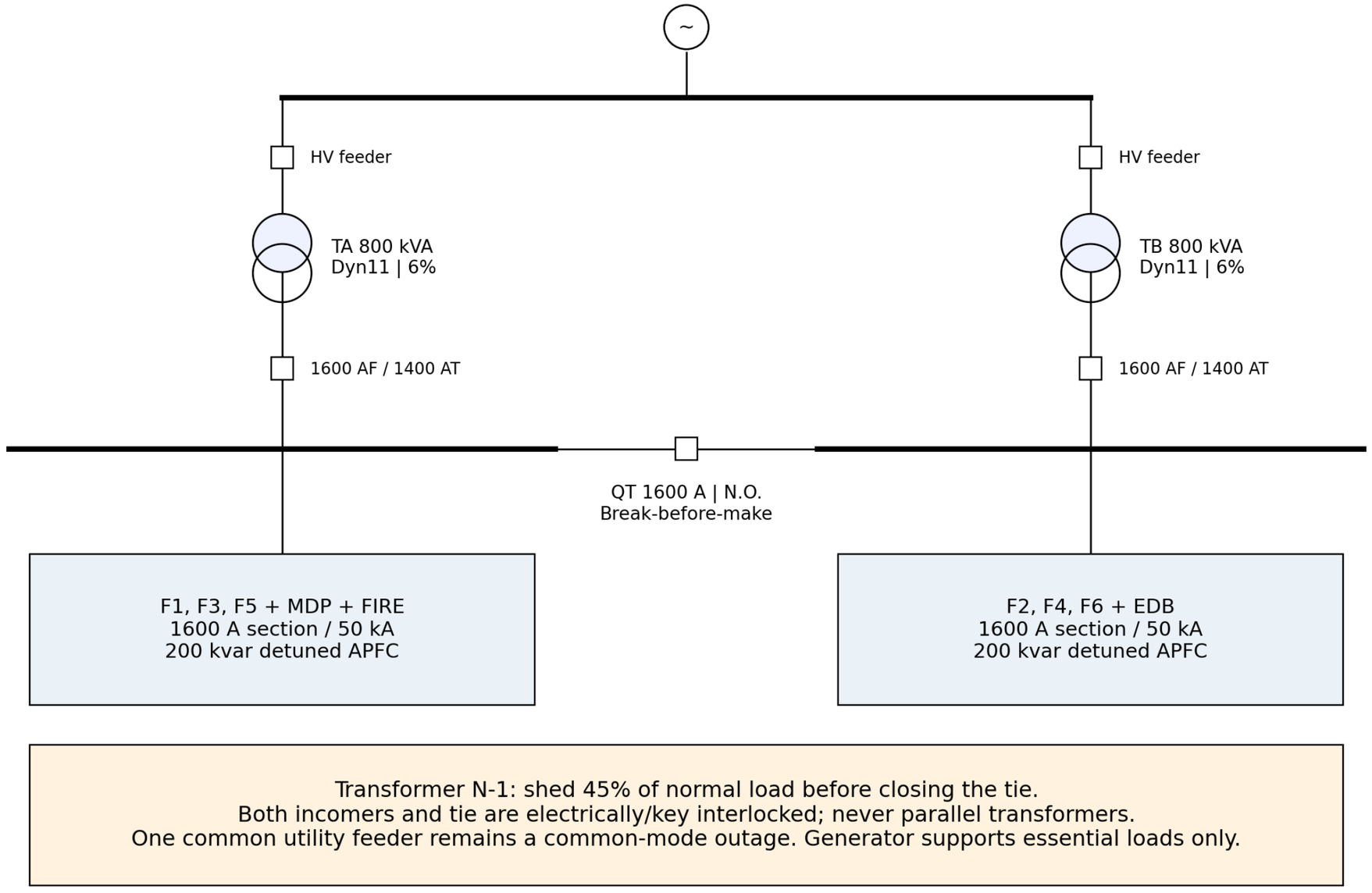
Alternative 2 - N-1 limitation and correction

```
Full future load on one 800 kVA transformer:
Winding current loading=131.109% >100%: NOT OK
Corrective action: retain 55% normal coincident load, shed 45%
Pretained=.55×1018.7364=560.3050 kW
Qretained=.55×540.8170=297.4494 kVAr (before APFC)
Solved surviving-transformer loading=70.894%
With harmonic capacity allowance=74.625%: OK
Conservative tie drop, all retained power through 10 m /6×300 mm²
ΔVtie≈(568.5812×1000×0.0001317
+119.1945×1000×0.0001600)/400=0.2348 V
Corrected minimum after tie and 1% branch reserve
=401.8852-0.2348-4=397.6504 V >380: OK
```

The N-1 solver collapses both sections onto the surviving transformer and conservatively uses the same main cable route; the added tie drop above bounds the omitted 10 m tie. A retained-load list prioritizes EDB, duty water pumping, one lift and selected floor circuits. The 55% fraction is an operating limit, not permission to interrupt fire-pump service. During fire duty the separate generator can support the fire circuit and essentials. Break-before-make transfer and load-shedding permissives must be hardwired.

The HV feeder remains a common failure point; neither alternative provides utility feeder N-1 redundancy. A2 improves transformer-maintenance flexibility and partial normal-load continuity only. Future feeder upgrades follow A1; the two units give more aggregate normal capacity but not full future N-1 capacity.

Alternative 2 | split bus, normally open tie



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Figure 8. E-08: alternative split-bus configuration and N-1 load-shedding constraint.

14 Technical-Economic Comparison

Table 26. Required design-alternative comparison

Criterion	Alternative 1	Alternative 2
Initial Cost	PHP 28,980,670	PHP 33,953,667
Transformer Capacity	1 ×1250 kVA; 1187.5 kVA planning thermal capability	2 ×800 kVA; 1520 kVA total planning capability
Feeder Capacity	Main: 10×300 mm²/phase, Iz 2777.32 A; Ir 2200 A	Each main/tie: 6×300 mm²/phase, Iz 1666.39 A; Ir 1400 A
Maximum Loading	Normal Y10: 83.12%; low-source fire: 88.67%	Normal Y10: 68.30%; N-1 full: 131.11% (shed required)
Minimum Voltage	Y10 low source panel: 388.92 V; load reserve: 384.92 V	Y10 low source panel: 389.90 V; load reserve: 385.90 V
Short-Circuit Requirement	Max LV bound 38.09 kA; select 50 kA	Max LV bound 26.91 kA; select 50 kA
Future Capacity	Meets year 10; close review at end of horizon	Higher normal spare capacity; N-1 retains 55% demand
Reliability	One transformer; essential standby only	Transformer outage permits reduced service with tie
Overall Assessment	Recommended for lower cost and simpler operation	Viable where partial normal-load continuity merits extra cost

Both estimates exceed the PHP 25 million soft planning target: A1 exceeds it by PHP 3,980,670 (15.9%); A2 by PHP 8,953,667. **Budget status: NOT OK against target.** Technical feasibility is maintained; procurement requires a revised funding allowance or value engineering supported by verified vendor prices and installation factors. No required cable protection is deleted to force the estimate below budget.

Budget estimate - common equipment and conductors

All unit prices below are **Assumed designer budget allowances in PHP, including VAT**, at October 2026 price basis. They are not quotations, market observations or a tender estimate. Installation and contingency are additional. End-use appliances, general branch wiring, utility contribution, civil/architectural works and land are excluded.

Table 27. Calculated common bill of materials

Tag / description	Qty	Unit PHP	Amount PHP
HV-01 - 17.5 kV 630 A 25 kA VCB/RMU, relays, CTs/VTs, meter provision	1	1,700,000	1,700,000
HV-C - 3 x 1C 50 mm2 shielded 15 kV cable, 20 m route, terminations included	1	240,000	240,000
P-F - 400 A 50 kA floor panel assemblies	6	145,000	870,000
P-E - 400 A essential panel, 320 A feeder, 50 kA	1	170,000	170,000
P-M - 160 A frame mechanical panel, 125 A trip, 50 kA	1	110,000	110,000
C-F - 185 mm2 Cu XLPE phase and 200% neutral cores, 2 sets x 5 cores x 300 m total routes	3,000	1,800	5,400,000
PE-F - 120 mm2 Cu floor PE, 2 conductors x 300 m	600	1,120	672,000
C-E - 150 mm2 Cu phase/200% neutral EDB, 2 sets x 5 cores x 30 m	300	1,420	426,000
PE-E - 95 mm2 Cu EDB PE, 2 x 30 m	60	900	54,000
C-M - 95 mm2 Cu mechanical 3P+N, 4 x 60 m	240	900	216,000
PE-M - 95 mm2 Cu mechanical PE, 60 m	60	900	54,000
FP-C - 120 mm2 Cu fire 3P x 35 m	105	1,120	117,600
FP-PE - 70 mm2 Cu fire PE x 35 m	35	670	23,450
FP-ATS - 200 A listed fire-pump controller/ATS, 50 kA, locked-rotor rated	1	650,000	650,000
GEN - 400 kVA / 320 kW diesel generator, 400 V, 60 Hz, installed accessories	1	2,500,000	2,500,000
E-ATS - 400 A four-pole essential ATS, conditional 50 kA withstand	1	350,000	350,000
GEN-C - Generator/essential transfer cabling, 3 sets 240 mm2 + PE, transfer subfeeders	1	900,000	900,000
EARTH - 300 mm2 main bond, earth ring, 12 rods and inspection pits allowance	1	280,000	280,000
TRAY - Separated cable tray tiers, supports, firestopping and spare capacity	1	500,000	500,000
TEST - Protection tests, PF/harmonic survey, commissioning	1	180,000	180,000

Budget estimate - alternative-specific equipment and totals

Table 28. A1 additional materials

Description	Qty	Unit PHP	Amount PHP
1250 kVA Dyn11 6% transformer, taps, 49/87T protection sensors	1	2,100,000	2,100,000
2500 A MDB, 2500 AF/2200 AT ACB, 50 kA, 105 kA peak	1	1,100,000	1,100,000
300 mm2 phase/200% neutral: 10 sets x 5 cores x 15 m	750	2,850	2,137,500
150 mm2 Cu PE: 10 x 15 m	150	1,420	213,000
400 kvar NET 400 V, 16 x 25 kvar, 7% reactors, complete feeder	1	950,000	950,000

```
A1: each amount = quantity × unit price
Materials = PHP 21,913,550.00
Installation = .15×21913550.00 = PHP 3,287,032.50
Contingency = .15×(21913550.00+3287032.50)
= PHP 3,780,087.38
Total = 21913550.00+3287032.50+3780087.38
= PHP 28,980,669.88
±30% range: PHP 20,286,469 to 37,674,871
```

Table 29. A2 additional materials

Description	Qty	Unit PHP	Amount PHP
800 kVA Dyn11 6% transformer	2	1,550,000	3,100,000
1600 A sections with 1600 AF/1400 AT incomers and 1600 A tie, 50 kA	1	1,900,000	1,900,000
300 mm2: 2 x 6 sets x 5 cores x 15 m plus tie 6 x 5 x 10 m	1200	2,850	3,420,000
150 mm2: (2 x 6 x 15 m) + (6 x 10 m)	240	1,420	340,800
200 kvar net bank: 16 x 12.5 kvar, 7% reactors, complete feeder	2	550,000	1,100,000
Second transformer HV feeder/relay provision	1	400,000	400,000

```
A2: each amount = quantity × unit price
Materials = PHP 25,673,850.00
Installation = .15×25673850.00 = PHP 3,851,077.50
Contingency = .15×(25673850.00+3851077.50)
= PHP 4,428,739.12
Total = 25673850.00+3851077.50+4428739.12
= PHP 33,953,666.62
±30% range: PHP 23,767,567 to 44,139,767
```

Life-cycle screening and weighted decision

A simple comparison uses year-0 corrected losses repeated over 20 years, the assumed daily loss/load factors, PHP 12/kWh, 1% of material cost annual maintenance, and 8% real discount. This is a screening comparison, not a forecast with escalation or a discounted growth dispatch model. Identical generator operation/fuel is excluded from the difference because both alternatives use the same essential system.

```
A1: Eannual = core kW×8760 + peak variable kW×8760×.439375
+ bank nominal kvar×.005×8760×.5875
= 50262.943 kWh/year
Annual losses cost =50262.943×12=PHP 603,155.31
Maintenance=.01×materials=PHP 219,135.50/year
PV factor =[1-(1.08)^(-20)]/.08=9.818147
Life-cycle PV=capital+(loss cost+maintenance)×PV factor
=PHP 37,054,042.28

A2: Eannual = core kW×8760 + peak variable kW×8760×.439375
+ bank nominal kvar×.005×8760×.5875
= 54345.188 kWh/year
Annual losses cost =54345.188×12=PHP 652,142.26
Maintenance=.01×materials=PHP 256,738.50/year
PV factor =[1-(1.08)^(-20)]/.08=9.818147
Life-cycle PV=capital+(loss cost+maintenance)×PV factor
=PHP 42,877,191.92
```

Table 30. Weighted decision matrix - assumed weights and scores, 5 is best

Criterion	Weight	A1 score	A2 score	A1 weighted	A2 weighted
Capital cost	35%	5	3	1.75	1.05
Normal capacity and voltage	20%	4	5	0.80	1.00
Transformer outage continuity	15%	2	4	0.30	0.60
Operation and maintenance	15%	5	3	0.75	0.45
Expansion flexibility	15%	3	5	0.45	0.75
TOTAL	100%	-	-	4.05	3.85

A1 scores 4.05/5 versus A2 3.85/5 and costs PHP 4,972,997 less. The outcome is sensitive to how the owner values partial normal-service continuity: shifting 10 percentage points of weight from capital cost to transformer-outage continuity reverses the ranking. No outage-cost benefit is monetized without reliability and tenant interruption data.

15 Final Recommended Design

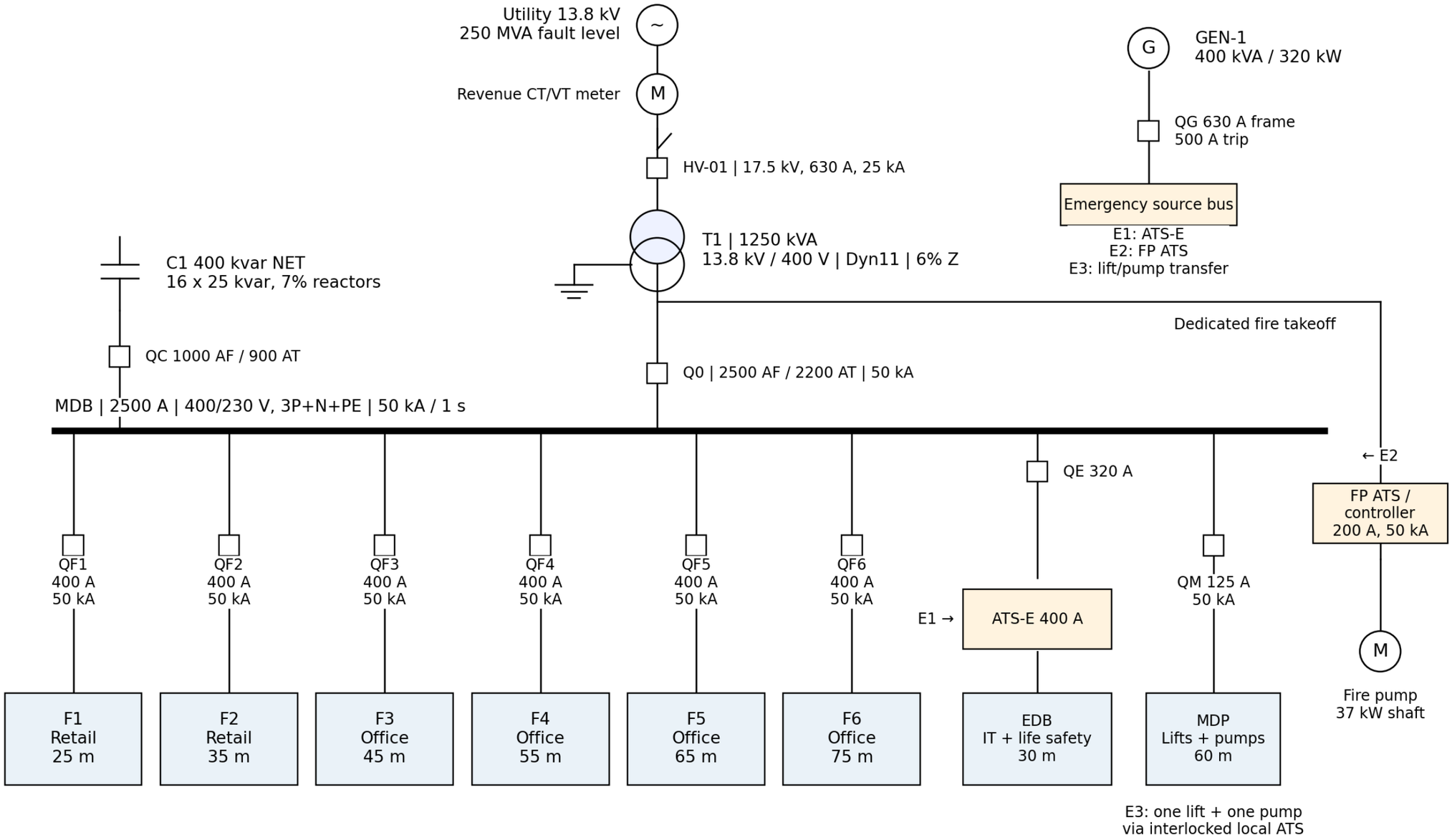
Recommend Alternative 1: one 1,250 kVA transformer with the specified derated parallel copper feeders, a 2,500 A MDB, 50 kA LV assemblies, 400 kVAr detuned APFC, and a 400 kVA essential generator. The choice meets the normal ten-year electrical criteria with less capital and operational complexity than A2. Estimated initial cost is PHP 28,980,670; this exceeds the soft budget target and is an explicit funding decision.

Table 31. Final rating register - governing selections

Tag	Final selection	Governing evidence
T1	1250 kVA, Dyn11, 13.8 kV/400 V, 6% Z, equivalent boost 1.025	Future demand +10% margin, 0.95 capability factor
HV-01	17.5 kV, 630 A, 25 kA; metering and protection CT/VT	250 MVA utility source + bounded motor contribution
Q0/MDB	2500 AF/2200 AT; 2500 A bus; 50 kA Icu/Ics/Icw 1 s; ≥105 kA peak	Future thermal and maximum LV fault screen
MAIN	10×300 mm² per phase; 20×300 mm² N; 10×150 mm² PE; 15 m	Iz=2777.32 A, Ir=2200 A
F1-F6	400 A /50 kA; 2×185 mm² phase, 4×185 N, 2×120 PE	Future 380 V RMS/125% inequality
EDB	320 AT /400 AF, 50 kA; 2×150 mm² phase, 4×150 N, 2×95 PE	Essential normal feeder
MDP	125 AT /160 AF, 50 kA; 1×95 mm² phase/N/PE	Future ampacity and PE adiabatic check
FIRE	200 A listed controller/ATS, 50 kA; 3×120 mm² phase +70 PE	37 kW shaft; 69.81 A calculated running current
C1/QC	400 kVAr NET; 16×25; 7%, 60 Hz; QC 1000 AF/900 AT; 4×240 phase +4×120 PE	PCC target, V², full-bank current and resonance
GEN-1/QG	400 kVA/320 kW; QG 630 AF/500 AT; 3×240 phase/N +3×120 PE	Essential running demand and starting screen
ATS-E	400 A 4-pole; conditional withstand 50 kA	Normal/standby isolation and neutral switching

The plant footprint reserves future transformer capacity; it does not assume parallel connection. The final SLD is the controlling topology, and the rating register resolves abbreviated drawing labels. Procurement must preserve the assumed installation arrangement and rated fault withstand.

Final single-line diagram | Alternative 1



Legend: ~ utility | overlapping circles transformer | □ breaker | M meter/motor (labelled) | G generator

All LV switchgear / ATS conditional withstand: 50 kA. E1-E3 are matched emergency-source connections; no utility parallel.

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Figure 9. E-02: final recommended single-line diagram; QC is labeled by both frame and trip rating.

16 Conclusions and Recommendations

The assumed commercial building requires 758.04 kW and 402.42 kVAr at normal peak, or 858.23 kVA at PF 0.8833. At 3% annual growth, year-10 normal demand is 1018.74 kW. The single 1,250 kVA alternative supplies it with transformer current loading of 83.12% at nominal supply and corrected PF 0.9737.

All selected normal floor/area cables pass the explicit future-load inequality. The worst low-source year-10 panel voltage is 388.92 V, leaving 384.92 V after the assumed branch allowance. The maximum LV fault bound is 38.09 kA, supporting the selected 50 kA equipment. No exact selective-coordination claim is made without device curves.

The 400 kVAr automatic detuned bank improves supply PF, current and voltage. Its operating steps depend on transformer VAR demand and bus voltage. The analysis finds no credible present energy-only payback under the frozen price/loss assumptions. The two-transformer alternative is feasible for normal service and reduced-load transformer N-1; it cannot carry full future demand on one 800 kVA unit.

Actions before procurement and commissioning

Table 32. Required closure actions

Priority	Action / acceptance
Utility	Obtain actual maximum/minimum fault MVA, X/R, high/low voltage, grounding and metering requirements. Recheck the fixed tap, including no-load high voltage.
Cables	Confirm PEC load/ampacity and terminal ratings, tray grouping, neutral terminations and manufacturer R/X; preserve equal parallel lengths.
Transformer	Verify loss guarantees, harmonic capability, actual tap ratio, %Z tolerance, 87T/REF and thermal curves.
Protection	Obtain breaker TCC, I^2t, ATS conditional ratings, motor starts and generator decrement curve; verify coordination and earth-fault protection.
Power quality	Measure phase currents, THDi, spectrum and PCC PF; perform detuned-bank frequency scan and stage tests.
Life safety	Validate listed fire-pump controller/transfer arrangement and generator transient acceleration with manufacturer data.
Budget	Secure funding for the stated estimate or value-engineer from quotations; do not reduce protective ratings without recalculation.

These closure actions identify the limits of assumed academic inputs; they do not replace the completed calculations. The independent solver package permits revised verified inputs to be studied without changing the documented Revision 0 brief.

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18 Appendices

Appendix A - Simulation Results (Independent Solver)

The delivered solver.py was executed with NumPy in this workspace. It models balanced positive-sequence constant-power loads on a radial network, source and transformer series impedances, transformer no-load loss, capacitor assembly real loss, and capacitor injection proportional to $|V|^2$. No ETAP, PowerWorld or MATLAB run is claimed.

```
Per phase: Sphase=(P+jQ)/3
Load current Ik=conj(Sphase/Vk)
Capacitor current Icap=+j Qcap,nom Vk/(3 Vphase,nom²)
Backward sweep: Ibranch=Σ downstream load/cap currents
Forward sweep: Vk=Vparent-Zbranch Ibranch
Stop when max|Vk,new-Vk,old|<10⁻⁹ V
Branch loss =3|Ibranch|²(R+jX)
Fault check: build Ybus; Zbus=Ybus⁻¹
I″k=cVphase/|Zbus,kk| (utility-only component)
```

The phase-domain model refers the MV network to 400 V. The transformer tap is represented as an equivalent LV slack-source boost, without a separate tap-dependent source-impedance adjustment. This approximation is small for the stated tap and source stiffness. It does not model unbalanced load, saturation, transients, zero-sequence or harmonic frequency response. The APFC search tests every available step and selects the smallest nonleading combination meeting 0.97 PCC displacement PF.

Table 33. Executed-case convergence and power-balance validation

Case	Iterations	PCC kW	PCC kVAr	PCC PF
A1_Y0_uncorrected	9	768.51	440.05	0.86780
A1_Y0_corrected	10	768.38	177.58	0.97432
A1_Y10_uncorrected	10	1,036.57	610.37	0.86171
A1_Y10_corrected	11	1,035.58	242.34	0.97369
A1_Y10_low_utility	11	1,036.41	239.27	0.97437
A2_Y10_corrected	10	1,034.93	254.51	0.97107
A1_Y10_low_fire	11	1,078.51	245.68	0.97502
A2_Y10_N1_shed	10	568.58	119.19	0.97873

Validation: all cases converged; source real power minus assigned load minus network/core/capacitor losses is below 10^{-5} kW. Fault path sums and the independently assembled/inverted admittance matrix agree within numerical round-off. Full unrounded inputs/outputs are included in JSON and CSV.

Appendix A - final design load-flow buses

Table 34. Simulation Results (Independent Solver) - present/future corrected bus data

Bus	Y0 V	Y0 angle °	Y10 V	Y10 angle °
GRID	410.00	0.0000	410.00	0.0000
PCC	409.60	-0.1631	409.46	-0.2198
HV	409.59	-0.1631	409.44	-0.2198
TA	404.13	-2.0988	402.14	-2.8416
MDB-A	403.86	-2.1311	401.77	-2.8859
CAP-A	403.93	-2.1422	401.87	-2.9014
F1	403.23	-2.1436	400.92	-2.9030
F2	402.97	-2.1487	400.57	-2.9098
F3	402.79	-2.1534	400.32	-2.9163
F4	402.55	-2.1584	400.00	-2.9230
F5	402.31	-2.1634	399.68	-2.9298
F6	402.07	-2.1684	399.36	-2.9366
EDB	403.33	-2.1483	401.06	-2.9093
MDP	402.20	-2.0731	399.53	-2.8070
FIRE	404.13	-2.0988	402.14	-2.8416

GRID/PCC/HV voltages are displayed on the 400 V reference side. For physical PCC voltage, divide the reported LV-referred voltage by 1.025 and multiply by 13,800/400. TA, MDB and load buses are actual LV values. Transformer no-load loss is represented at its secondary node. FIRE is unloaded in these normal cases.

Appendix A - final design branch flow and losses

Table 35. Simulation Results (Independent Solver) - year 10 corrected branches

Branch	Current A	kW sent	kVAr sent	Loading %	Loss kW	Loss kVAr
SOURCE	1,499.65	1,036.01	246.64	0.00	0.4297	4.2966
HV-CABLE	1,499.65	1,035.58	242.34	0.00	0.0560	0.0136
TR-A	1,499.65	1,035.53	242.33	83.12	8.6360	51.0913
MAIN-A	1,497.11	1,025.09	191.24	53.90	0.7968	0.9683
CAPFEED-A	507.55	1.85	-353.19	52.60	0.0954	0.0927
F1	248.17	151.26	83.34	60.79	0.2959	0.2217
F2	248.38	151.38	83.43	60.85	0.4149	0.3109
F3	233.85	143.04	77.61	57.29	0.4729	0.3544
F4	234.04	143.14	77.69	57.33	0.5789	0.4338
F5	234.23	143.25	77.77	57.38	0.6853	0.5135
F6	234.42	143.35	77.85	57.42	0.7920	0.5935
EDB	150.70	99.61	32.79	42.10	0.1615	0.0981
MDP	83.02	47.42	33.00	62.20	0.3095	0.1191
FIRE	0.00	0.00	-0.00	0.00	0.0000	0.0000

```
Year10 PCC power balance:
Pload=1018.736399 kW
Pnetwork+core+APFC=16.845050 kW
PCC=1035.581449 kW
Residual=-0.000000000 kW: OK
```

Source and HV cable currents in this table are referred to LV. Physical HV current equals the tabulated referred current ×400/13,800, subject to the stated equivalent tap approximation. Source utility impedance loss is outside the PCC/customer loss total. Transformer core and APFC real losses are separate nodal loads and therefore not inside branch I^2R columns.

Appendix B - Calculated vs. Simulated

Table 36. Calculated vs. Simulated - voltage

Quantity	Calculated	Simulated	Difference %	Explanation
TA V	402.6275	402.1385	0.1216	Fixed 400 V hand current vs iterative constant-PQ / V² cap
MDB-A V	402.2565	401.7681	0.1216	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F1 V	401.4028	400.9163	0.1213	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F2 V	401.0613	400.5746	0.1215	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F3 V	400.8120	400.3247	0.1217	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F4 V	400.4910	400.0026	0.1221	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F5 V	400.1700	399.6798	0.1226	Fixed 400 V hand current vs iterative constant-PQ / V² cap
F6 V	399.8490	399.3566	0.1233	Fixed 400 V hand current vs iterative constant-PQ / V² cap
EDB V	401.5496	401.0630	0.1213	Fixed 400 V hand current vs iterative constant-PQ / V² cap
MDP V	400.0200	399.5286	0.1230	Fixed 400 V hand current vs iterative constant-PQ / V² cap

Appendix B - independent fault comparison

Table 37. Calculated vs. Simulated - faults

Quantity	Calculated	Simulated	Difference %	Explanation
PCC utility kA	10.459244	10.459244	1.87e-13	Series Z sum vs independent Ybus inversion
HV utility kA	10.416656	10.416656	2.05e-13	Series Z sum vs independent Ybus inversion
T utility kA	33.071783	33.071783	1.72e-13	Series Z sum vs independent Ybus inversion
MDB utility kA	32.412342	32.412342	1.75e-13	Series Z sum vs independent Ybus inversion
F1 utility kA	27.416292	27.416292	2.59e-13	Series Z sum vs independent Ybus inversion
F2 utility kA	25.767684	25.767684	2.34e-13	Series Z sum vs independent Ybus inversion
F3 utility kA	24.284746	24.284746	2.34e-13	Series Z sum vs independent Ybus inversion
F4 utility kA	22.947253	22.947253	2.63e-13	Series Z sum vs independent Ybus inversion
F5 utility kA	21.737322	21.737322	2.45e-13	Series Z sum vs independent Ybus inversion
F6 utility kA	20.639344	20.639344	2.07e-13	Series Z sum vs independent Ybus inversion
EDB utility kA	26.300269	26.300269	2.43e-13	Series Z sum vs independent Ybus inversion
MDP utility kA	13.961628	13.961628	2.42e-13	Series Z sum vs independent Ybus inversion
FIRE utility kA	20.146247	20.146247	2.47e-13	Series Z sum vs independent Ybus inversion

```
Percent difference =100×|calculated-simulated|/|simulated|
Identical fault results verify numerical implementation, not uncertain inputs.
The separate arithmetic motor bound is added equally after either method.
```

Appendix B - hand voltage path details

Every branch below uses ΔV=(P×1000 R+Q×1000 X)/400 with the summed downstream hand P,Q and nominal switched capacitor output. Receiving voltage equals parent voltage minus ΔV. These values deliberately do not reuse the nonlinear solution.

```
SOURCE: P=1022.286399 kW; Q=190.817019 kVAr
ΔV=(1022.286399×1000×0.000063682
+190.817019×1000×0.000636824)/400 = 0.466546 V
Vreceiving=410.000000-0.466546=409.533454 V

HV-CABLE: P=1022.286399 kW; Q=190.817019 kVAr
ΔV=(1022.286399×1000×0.000008301
+190.817019×1000×0.000002016)/400 = 0.022176 V
Vreceiving=409.533454-0.022176=409.511278 V

TR-A: P=1022.286399 kW; Q=190.817019 kVAr
ΔV=(1022.286399×1000×0.001280000
+190.817019×1000×0.007572582)/400 = 6.883760 V
Vreceiving=409.511278-6.883760=402.627517 V

MAIN-A: P=1020.486399 kW; Q=190.817019 kVAr
ΔV=(1020.486399×1000×0.000118500
+190.817019×1000×0.000144000)/400 = 0.371013 V
Vreceiving=402.627517-0.371013=402.256504 V

CAPFEED-A: P=1.750000 kW; Q=-350.000000 kVAr
ΔV=(1.750000×1000×0.000123437
+-350.000000×1000×0.000120000)/400 = -0.104460 V
Vreceiving=402.256504--0.104460=402.360964 V

F1: P=150.962127 kW; Q=83.116123 kVAr
ΔV=(150.962127×1000×0.001601351
+83.116123×1000×0.001200000)/400 = 0.853707 V
Vreceiving=402.256504-0.853707=401.402797 V

F2: P=150.962127 kW; Q=83.116123 kVAr
ΔV=(150.962127×1000×0.002241892
+83.116123×1000×0.001680000)/400 = 1.195190 V
Vreceiving=402.256504-1.195190=401.061314 V

F3: P=142.562650 kW; Q=77.253252 kVAr
ΔV=(142.562650×1000×0.002882432
+77.253252×1000×0.002160000)/400 = 1.444486 V
Vreceiving=402.256504-1.444486=400.812018 V

F4: P=142.562650 kW; Q=77.253252 kVAr
ΔV=(142.562650×1000×0.003522973
+77.253252×1000×0.002640000)/400 = 1.765482 V
Vreceiving=402.256504-1.765482=400.491022 V

F5: P=142.562650 kW; Q=77.253252 kVAr
ΔV=(142.562650×1000×0.004163514
+77.253252×1000×0.003120000)/400 = 2.086479 V
Vreceiving=402.256504-2.086479=400.170025 V

F6: P=142.562650 kW; Q=77.253252 kVAr
ΔV=(142.562650×1000×0.004804054
+77.253252×1000×0.003600000)/400 = 2.407476 V
Vreceiving=402.256504-2.407476=399.849028 V

EDB: P=99.449812 kW; Q=32.687572 kVAr
ΔV=(99.449812×1000×0.002370000
+32.687572×1000×0.001440000)/400 = 0.706915 V
Vreceiving=402.256504-0.706915=401.549589 V
```

```
MDP: P=47.111735 kW; Q=32.884194 kVAr
ΔV=(47.111735×1000×0.014968421
+32.884194×1000×0.005760000)/400 = 2.236503 V
Vreceiving=402.256504-2.236503=400.020001 V

FIRE: P=0.000000 kW; Q=0.000000 kVAr
ΔV=(0.000000×1000×0.006912500
+0.000000×1000×0.003360000)/400 = 0.000000 V
Vreceiving=402.627517-0.000000=402.627517 V
```

Appendix C - full fault substitutions at every bus

A1 maximum-fault calculation register

```
PCC: ZΣ=0.000070051+j0.000700506 Ω (LV base)
Iutility=1.10×400/(√3×0.000704000)/1000×0.02898551
=10.459244 kA
Idesign=10.459244+0.145598=10.604842 kA: OK

HV: ZΣ=0.000078351+j0.000702523 Ω (LV base)
Iutility=1.10×400/(√3×0.000706878)/1000×0.02898551
=10.416656 kA
Idesign=10.416656+0.145598=10.562254 kA: OK

T: ZΣ=0.001240924+j0.007580396 Ω (LV base)
Iutility=1.10×400/(√3×0.007681295)/1000×1.00000000
=33.071783 kA
Idesign=33.071783+5.023145=38.094928 kA: OK

MDB: ZΣ=0.001327129+j0.007724396 Ω (LV base)
Iutility=1.10×400/(√3×0.007837574)/1000×1.00000000
=32.412342 kA
Idesign=32.412342+5.023145=37.435487 kA: OK

F1: ZΣ=0.002492062+j0.008924396 Ω (LV base)
Iutility=1.10×400/(√3×0.009265809)/1000×1.00000000
=27.416292 kA
Idesign=27.416292+5.023145=32.439437 kA: OK

F2: ZΣ=0.002958035+j0.009404396 Ω (LV base)
Iutility=1.10×400/(√3×0.009858632)/1000×1.00000000
=25.767684 kA
Idesign=25.767684+5.023145=30.790829 kA: OK

F3: ZΣ=0.003424008+j0.009884396 Ω (LV base)
Iutility=1.10×400/(√3×0.010460646)/1000×1.00000000
=24.284746 kA
Idesign=24.284746+5.023145=29.307891 kA: OK

F4: ZΣ=0.003889981+j0.010364396 Ω (LV base)
Iutility=1.10×400/(√3×0.011070350)/1000×1.00000000
=22.947253 kA
Idesign=22.947253+5.023145=27.970398 kA: OK

F5: ZΣ=0.004355953+j0.010844396 Ω (LV base)
Iutility=1.10×400/(√3×0.011686541)/1000×1.00000000
=21.737322 kA
Idesign=21.737322+5.023145=26.760468 kA: OK

F6: ZΣ=0.004821926+j0.011324396 Ω (LV base)
Iutility=1.10×400/(√3×0.012308246)/1000×1.00000000
=20.639344 kA
Idesign=20.639344+5.023145=25.662489 kA: OK

EDB: ZΣ=0.003051229+j0.009164396 Ω (LV base)
Iutility=1.10×400/(√3×0.009658993)/1000×1.00000000
=26.300269 kA
Idesign=26.300269+5.023145=31.323414 kA: OK

MDP: ZΣ=0.012216182+j0.013484396 Ω (LV base)
Iutility=1.10×400/(√3×0.018195165)/1000×1.00000000
=13.961628 kA
Idesign=13.961628+5.023145=18.984773 kA: OK
```

```
FIRE: ZΣ=0.006269549+j0.010940396 Ω (LV base)
Iutility=1.10×400/(√3×0.012609501)/1000×1.00000000
=20.146247 kA
Idesign=20.146247+5.023145=25.169393 kA: OK
```

A2 maximum-fault calculation register

```
PCC: ZΣ=0.000070051+j0.000700506 Ω (LV base)
Iutility=1.10×400/(√3×0.000704000)/1000×0.02898551
=10.459244 kA
Idesign=10.459244+0.145598=10.604842 kA: OK

HV: ZΣ=0.000078351+j0.000702523 Ω (LV base)
Iutility=1.10×400/(√3×0.000706878)/1000×0.02898551
=10.416656 kA
Idesign=10.416656+0.145598=10.562254 kA: OK

T: ZΣ=0.002076729+j0.011418015 Ω (LV base)
Iutility=1.10×400/(√3×0.011605338)/1000×1.00000000
=21.889421 kA
Idesign=21.889421+5.023145=26.912566 kA: OK

MDB: ZΣ=0.002220404+j0.011658015 Ω (LV base)
Iutility=1.10×400/(√3×0.011867582)/1000×1.00000000
=21.405719 kA
Idesign=21.405719+5.023145=26.428864 kA: OK

F1: ZΣ=0.003385337+j0.012858015 Ω (LV base)
Iutility=1.10×400/(√3×0.013296204)/1000×1.00000000
=19.105763 kA
Idesign=19.105763+5.023145=24.128908 kA: OK

F2: ZΣ=0.003851310+j0.013338015 Ω (LV base)
Iutility=1.10×400/(√3×0.013882911)/1000×1.00000000
=18.298332 kA
Idesign=18.298332+5.023145=23.321478 kA: OK

F3: ZΣ=0.004317283+j0.013818015 Ω (LV base)
Iutility=1.10×400/(√3×0.014476756)/1000×1.00000000
=17.547724 kA
Idesign=17.547724+5.023145=22.570870 kA: OK

F4: ZΣ=0.004783255+j0.014298015 Ω (LV base)
Iutility=1.10×400/(√3×0.015076895)/1000×1.00000000
=16.849233 kA
Idesign=16.849233+5.023145=21.872379 kA: OK

F5: ZΣ=0.005249228+j0.014778015 Ω (LV base)
Iutility=1.10×400/(√3×0.015682605)/1000×1.00000000
=16.198464 kA
Idesign=16.198464+5.023145=21.221609 kA: OK

F6: ZΣ=0.005715201+j0.015258015 Ω (LV base)
Iutility=1.10×400/(√3×0.016293267)/1000×1.00000000
=15.591356 kA
Idesign=15.591356+5.023145=20.614501 kA: OK

EDB: ZΣ=0.003944504+j0.013098015 Ω (LV base)
Iutility=1.10×400/(√3×0.013679075)/1000×1.00000000
=18.571001 kA
Idesign=18.571001+5.023145=23.594146 kA: OK

MDP: ZΣ=0.013109457+j0.017418015 Ω (LV base)
Iutility=1.10×400/(√3×0.021800117)/1000×1.00000000
=11.652879 kA
Idesign=11.652879+5.023145=16.676024 kA: OK
```

```
FIRE: ZΣ=0.007105354+j0.014778015 Ω (LV base)
Iutility=1.10×400/(√3×0.016397432)/1000×1.00000000
=15.492311 kA
Idesign=15.492311+5.023145=20.515457 kA: OK
```

Appendix D - equipment quantities and installation notes

Table 38. Final equipment schedule

Tag	Description / key rating	Quantity
T1	1250 kVA transformer, 13.8/0.4 kV, Dyn11, 6% Z	1
HV-01	17.5 kV/630 A/25 kA switchgear including CT/VT and relay	1
MDB/Q0	2500 A main board, 2500 AF/2200 AT, 50 kA, 105 kA peak	1
QF/PF	400 A/50 kA floor panel with outgoing feeder breaker	6
QE/EDB	400 A board, 320 AT feeder, 50 kA	1
QM/MDP	160 AF/125 AT mechanical panel, 50 kA	1
FP-ATS	200 A listed fire controller/ATS, 50 kA	1
C1	400 kvar net, 16×25 kvar, 7% reactors, 480 V capacitor elements	1
QC	1000 AF/900 AT, 50 kA	1
CAP-STEP	25 kvar net, 63 A fuse, capacitor-duty contactor, reactor	16
GEN-1	400 kVA/320 kW, 400 V/60 Hz	1
QG	630 AF/500 AT generator feeder breaker	1
ATS-E	400 A, four-pole, open transition	1
ATS-L/W	100 A local essential motor transfer assemblies	2
EARTH	3 m copper-bonded rods with ring and pits	12
METER	PCC revenue CT/VT provision and customer PQ/PF transducer	1

Capacitor feeder, step controls and reactor assembly are included in the APFC budget line; local essential transfer equipment/cabling are included in the generator transfer allowance. Main cable quantities count five current-carrying conductors per set (three phase plus two neutrals), plus separate PE. Cable reserve length is not silently added; route and termination allowances are inside unit-price/installation uncertainty.

```
Floor phase+neutral cable = (25+35+45+55+65+75)×2 sets×5 cores
=300×2×5=3000 conductor-metres
Floor PE =300×2=600 conductor-metres
Main phase+neutral=15×10×5=750 conductor-metres
Main PE=15×10=150 conductor-metres
EDB phase+neutral=30×2×5=300 conductor-metres
MDP phase+neutral=60×4=240 conductor-metres
Fire phase=35×3=105 conductor-metres
```

Appendix E - model inputs for re-entry in other software

Table 39. Bus/load model - values at year 0

Bus	kW	kVAr	Parent	Cable input
F1	112.330000	61.846201	MDB-A	25 m; 2×185 mm²; R=0.0016014 Ω, X=0.0012000 Ω
F2	112.330000	61.846201	MDB-A	35 m; 2×185 mm²; R=0.0022419 Ω, X=0.0016800 Ω
F3	106.080000	57.483675	MDB-A	45 m; 2×185 mm²; R=0.0028824 Ω, X=0.0021600 Ω
F4	106.080000	57.483675	MDB-A	55 m; 2×185 mm²; R=0.0035230 Ω, X=0.0026400 Ω
F5	106.080000	57.483675	MDB-A	65 m; 2×185 mm²; R=0.0041635 Ω, X=0.0031200 Ω
F6	106.080000	57.483675	MDB-A	75 m; 2×185 mm²; R=0.0048041 Ω, X=0.0036000 Ω
EDB	74.000000	24.322624	MDB-A	30 m; 2×150 mm²; R=0.0023700 Ω, X=0.0014400 Ω
MDP	35.055556	24.468928	MDB-A	60 m; 1×95 mm²; R=0.0149684 Ω, X=0.0057600 Ω
FIRE	0.000000	0.000000	TA	35 m; 1×120 mm²; R=0.0069125 Ω, X=0.0033600 Ω

Use 100% constant-power loads; multiply every normal P,Q by 1.03^{n}. Fire case adds 41.111111 kW and 25.478648 kVAr at FIRE without normal growth. Set source 13.8 kV, nominal short-circuit strength 250 MVA, X/R 10; maximum-fault source uses c=1.10 consistently. Transformer: 1250 kVA, 6% Z, 1% R; equivalent source boost 1.025; copper loss 12.5 kW at rated current, core 1.8 kW.

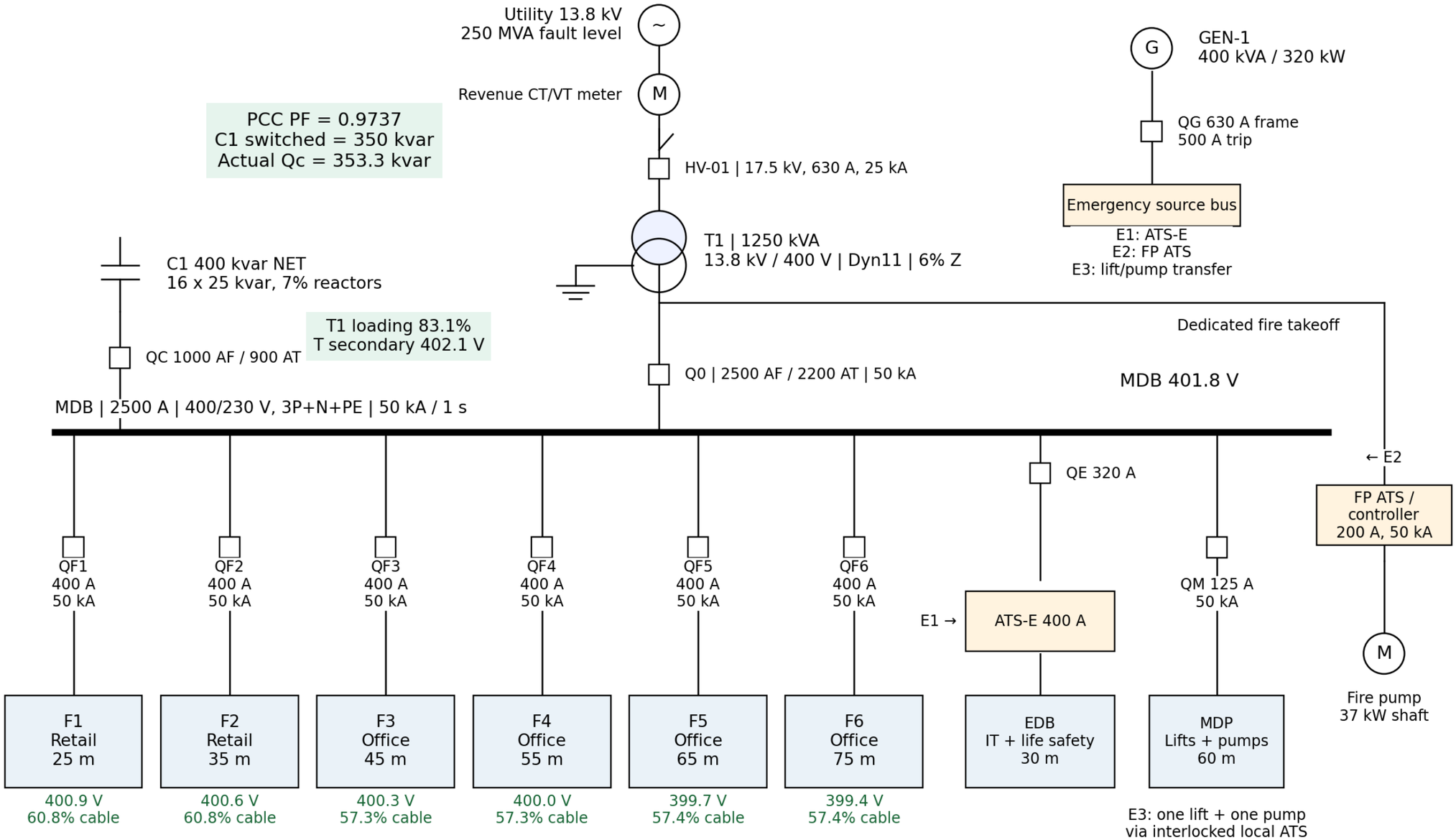
For the capacitor bank, use shunt susceptance corresponding to nominal switched kVAr at 400 V; do not use a constant negative-Q load if reproducing V^2 output. The delivered fundamental model represents reactor/capacitor net kVAr plus 5 W/kVAr real loss; it does not represent their frequency-dependent harmonic circuit. APFC states for each run are in calculated_results.json.

Table 40. Independent simulation package contents

File / pattern	Contents
design_inputs.json	Frozen numerical brief and load data; SHA-256 on Section 2
solver.py	Executable load-flow and nodal fault engine; reads inputs and writes results
analysis.py	Hand calculation, sizing, economic and comparison calculations
calculated_results.json	All cases, branch flows, voltages, faults and convergence
engineering_analysis.json	Sizing inequalities, hand voltages, budgets, PF losses and decision matrix
*_buses.csv / *_branches.csv	Unrounded case-specific load-flow outputs
*_faults.csv	A1/A2 maximum and minimum fault output tables
load_schedule.csv / feeder_inputs.csv / year_by_year.csv	Model allocation and growth inputs/results
README.txt	Run instructions, modeling limits and file map

Run `python solver.py` then `python analysis.py` in the simulation directory with NumPy installed. Figures and PDF are supplied separately; the full ZIP also contains the report/drawing generation scripts. Re-entry into ETAP is possible using these inputs, but the report makes no ETAP verification claim.

Independent solver model | Year 10 corrected



Legend: ~ utility | overlapping circles transformer | □ breaker | M meter/motor (labelled) | G generator

All LV switchgear / ATS conditional withstand: 50 kA. E1-E3 are matched emergency-source connections; no utility parallel.

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Prepared by John Ray O. Dalida | EEA108D-E06 | Group 1 | Mapua University

Figure 10. E-05: final design with year-10 solved bus voltages and equipment loading.

Appendix F - final quality checklist

1. Completeness
PASS: all eight tasks, two alternatives and the eighteen requested sections are present.

2. Rating consistency
PASS: final register, equipment quantities, BOM and drawings share the selected ratings. QC is explicitly 1000 AF /900 AT in the drawings and schedule.

3. Present/future electrical limits
PASS for stated normal and fire cases with APFC. No-tap/APFC-off failures and A2 full N-1 overload are identified with corrective actions. Zero-load high-tap voltage needs utility/tap closure.

4. Short-circuit equipment selection
PASS: calculated maximum LV duty below 50 kA, peak below 105 kA; HV below 25 kA. Ratings are standard procurement classes.

5. Power factor correction
PASS for fundamental sizing, stepping, reactor specification and benefits. Bare-bank resonance risk identified; harmonic-frequency scan remains a procurement validation. No invented positive payback.

6. Recommendation
PASS: A1 recommended at 4.05/5 weighted score and lower stated cost; funding gap against soft target is explicit.

7. Verification
PASS: executed independent load-flow solver and separately assembled nodal fault check; calculated/simulated errors explained.

8. Drawings and identity
PASS: six required drawings plus four supporting drawings; separately saved PNGs, title blocks, page-1 student header and every-page name/course footers.

Budget / construction status
OPEN: estimate exceeds PHP 25 million soft target; actual utility data, procurement curves, harmonic scan and site earthing remain commissioning/permit closure items.

End of Revision 0. All engineering judgments remain traceable to the frozen design brief, selected ratings and supplied calculation model.